PROGRESS REPORT Ⅰ

项目进度报告（一）

Tsinghua University, Beijing, 100084

University of Science and Technology Beijing, 100083

Feb, 2015

ABSTRACT

Vibration measurement and analysis are widely used in evaluating the working condition of machine. By measuring the elevator vibration especially car vibration and rope tension, the project aims at revealing the elevator vibration mechanism, excitation source, transfer rules and dynamic response which reflect the health condition. Three phases (modeling, measurement and analysis) are included in this project, while the progress and achievements of phase 1, i.e., dynamics modeling, are summarized in this report. Both a 17 DOF vertical vibration model and an 8 DOF horizontal vibration model are established, and the free vibration response and forced vibration response are achieved by solving the system of ordinary differential equations based on Matlab platform. Time domain, frequency domain and joint time and frequency domain analysis are used to reveal the frequency components constitution, which is of nonstationarity due to time-varying mass and stiffness. The model is also validated by comparison with in-situ measured vibration signals, and the results indicate that the model is correct and can be used in natural frequency prediction and fault feature detection.

The rest of this report is structured as follows. Sections 1 and 2 respectively introduces the project objects and research background. In section 3, the structure and parameters of project pilot elevator are exhibited. The vertical and horizontal models are set up in section 4, while their solution methods are explained in section 5. Free vibration analysis and forced vibration analysis are done in sections 6 and 7 respectively. Finally the conclusion and plans for the future work are suggested in section 8.

Key words: Elevator vibration, dynamics model, solution, frequency analysis

TABLE OF CONTENTS

ABSTRACT I
TABLE OF CONTENTS III
1 OBJECTS 1
2 RESEARCH BACKGROUND 3
3 PILOT ELEVATOR 5
3.1 Structure 5
3.2 Parameters 7
3.3 Explanation 8
4 DYNAMICS MODELLING 9
4.1 Vertical model 9
4.2 Horizontal model 13
5 EQUATION SOLUTION 15
5.1 ODE 15
5.2 Newmark-β 17
6 FREE VIBRATION ANALYSIS 18
6.1 Natural frequencies 18
6.1.1 Vertical natural frequencies 18
6.1.2 Horizontal natural frequencies 20
6.2 Vertical vibration analysis 20
6.3 Horizontal vibration analysis 24
6.4 Conclusion 26
7 FORCED VIBRATION ANALYSIS 27
7.1 External excitation 27
7.1.1 Vertical excitation 27
7.1.2 Horizontal excitation 28
7.2 Vertical vibration analysis 29
7.3 Horizontal vibration analysis 33
7.4 Conclusion 35
8 CONCLUSION AND FUTURE WORK 36
REFERENCE 37

1 OBJECTS

Elevators are of rapidly increasing significance along with the quick city development process. For elevator passengers, the safety as well as ride quality are of most concern. Vibration monitoring and diagnosis is widely used in machinery fault diagnosis, owing to its high accuracy and efficiency. To monitor the running condition of elevator and predict fault feature for avoiding accidents, elevator vibration measurement and analysis are of significance.

Aiming at ensuring the safety and comfortableness during elevator riding, the objective of the proposed project is to deepen the understanding of available vibration and force measurements thus provide reliable guidance for the ride quality monitoring of elevator. Both parties agree the proposed work will focus on the measurement and analysis of rope force as well as car vibration signals, thus revealing the running condition of the objective elevator.

Three main phases are included in the whole project, in which the vibration model building is undergoing. As agreed by both parties, reliable and simplified vibration model is necessary for theoretical derivation and numerical evaluation of meaningful information in the project, including each characteristic frequency, vibration indicators, etc. The complexity of elevator system's composition and mechanical structure indicate that deep study, appropriate simplification and innovative ideas in analysis are required in following tasks.

Task 1: Study of the mechanical structure, force distribution and dynamic behavior of the objective elevator, especially the vibration sources, transfer paths and existing vibration dampers. The key mechanical components should be highlighted while structures meaningless to vibration and smoothness can be simplified, after which the structure diagram for mathematical modeling is available. It is supported by KONE with appropriate elevator sample and carried out by Tsinghua.

Task 2: Vibration model building in both vertical and horizontal direction, including the determination of inherent parameters, element simplification, equations of mathematical physics selection and differential equation establishment. This mechanical model building requires the inherent parameters of the objective elevator to be available or calculable. Further validation of this model will be carried out during measurement and analysis in stage 2.

Task 3: Analysis of the established model in order to obtain characteristic quantities including each order of the natural frequencies, potential fault frequencies, required time or amplitude range, etc. It lays a foundation for the three aspects of common model analysis, i.e., vibration design, system identification and condition identification. The latter two aspects will be applied in the project, which focus on the vibration stimulation and response in both directions and help realize the goal of building bridges between available measurements and concealing vibration and tension laws.

According to the project timetable in the contract, the research progress is actually beyond expectation. By the end of February, the contents in phase 1 are all accomplished despite the missing of pilot elevator parameters. Both vertical and horizontal vibration models have been set up and preliminary validated. They have also been used to predict the vibration characteristics. Besides, part of phase 2 work have already been carried out, e.g., the two on-site tests. In general, the whole project is under good progress and much achievements have been obtained.

2 RESEARCH BACKGROUND

As a complex electromechanical system, elevator is of specific driving, transmission and moving components under the control of unitive controller. The vibration phenomena are inevitable due to manufacturing and/or assembling errors, as well as external excitations. In case of specific local or distributed fault, system unbalance and fierce vibration are easily generated and worth analysis. Usually the elevator vibration includes both vertical and horizontal case, and some researchers have made contributions to this field.

For elevator vertical vibration, which is stimulated by motor output torque and transferred mainly through hoist rope, have been under focus. Zhang and Mei [1] build a 10 DOF vertical vibration model based on Matlab to carry out vibration feature calculation. Zhen [2] etc., constructs an simple 1:1 elevator vertical and horizontal model, and further developed a dynamic analyzing software based on C programming language. A Mitsubishi elevator is also used for effectiveness evaluation. Wang [3] etc., studied 2:1 elevator trying to extract the frequency characteristic and inner relationship, with simulated excitation as well. Chen and Yu [4] focus on a TRJT1000/1.6-JXPC elevator in order to build a vertical 7 DOF vibration model. They also proposed an optimization method for kinetic characteristics analysis, together with Fortran program. However, they didn't take the time-variant parameters into consideration. Lei [5] studied the lab elevator and build a simulation model trying to adjust the controlling system for kinetic characteristics improvement. Gang [6] takes the mass of hoist rope and compensation rope into consideration, and use genetic algorithm to optimize the dynamic behavior based on elevator sensitivity. Zhang and Yang [7-8] discretize a 1:1 elevator 7DOF model and transform the time variant model into series of transient systems for obtaining numerical solution. Nai [9-10] study the key point of time-varying parameters modeling and build a 10DOF vertical model, in which the mass of hoist rope are discretized.

Research on horizontal vibration analysis are relatively less, because the force transmission path, horizontal structure and parameters are all related to the vibration response and are of complex mechanism. Feng and Zhang [11] systematically study the active control of high speed elevator horizontal vibration and proposed an active guide shoe based on hydraulic control. Both simulation and on-site test are applied to validate

the effectiveness of active guide shoe and controlling strategy. Robust control rules are also applied to reduce the horizontal vibration. Fu [12] regards the connection between car frame and car as rigid, and further build a horizontal 5DOF model using transformation of coordinates. Different guide rail excitations are added to obtain possible responses. But the horizontal damping cannot be ignored in real world case. Zhang [13] studied the influence of guide rail unevenness and running speed on horizontal vibration using a 12 DOF nonlinear model. Zhu [14] established a stochastic volatility model with fixed/kinetic hoist rope and corresponding vibration equations and boundary values, which lays a foundation for the study of relationship between hoist rope and car vibration.

From aforementioned literature, we can find that although vast amounts of work have been done in elevator vibration analysis field, most of them stay on the theoretical simulation stage and the factors having been taken into consideration are not complete. In vertical vibration analysis, the mass of hoist rope ought to be added and they are time variant. Guide rail excitation in horizontal direction always features randomness and are sensitive to fault. The elevator vertical and horizontal vibration demand for deep and complete research, together with more accurate model and on-site test.

3 PILOT ELEVATOR

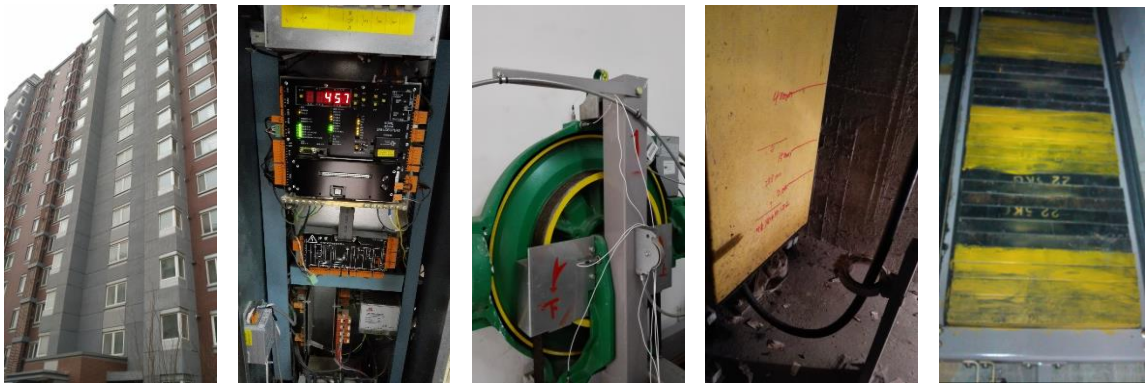
A KONE 3000X MiniSpace elevator (36594114) is chosen as the project pilot elevator. Project team managed to find out the main information of the structure and condition of this elevator from on-site test and inquiry. Basic parameters like the travel route, mass and mechanical parameters are estimated by field measurement and experience. In this section, the information for dynamics modelling are exhibited.

3.1 Structure

Mounted in No.13 building of GuoShiHui community in 2012, the elevator has run for about 450 thousand times. Since the average life is 15 years, this elevator is quite new. The traction ratio is 2:1, thus the main structural components include:

Traction motor: The main power source
Car: The passenger direct contact component
Car frame: The direct hoist component
Counterweight: The mass balance part
Hoist rope: For traction
Diversion sheave:
Pulley: For car frame traction
Compensation rope: The mass balance rope

Figure 1 shows some photos and a sketch illustrates the actual structure of this elevator.



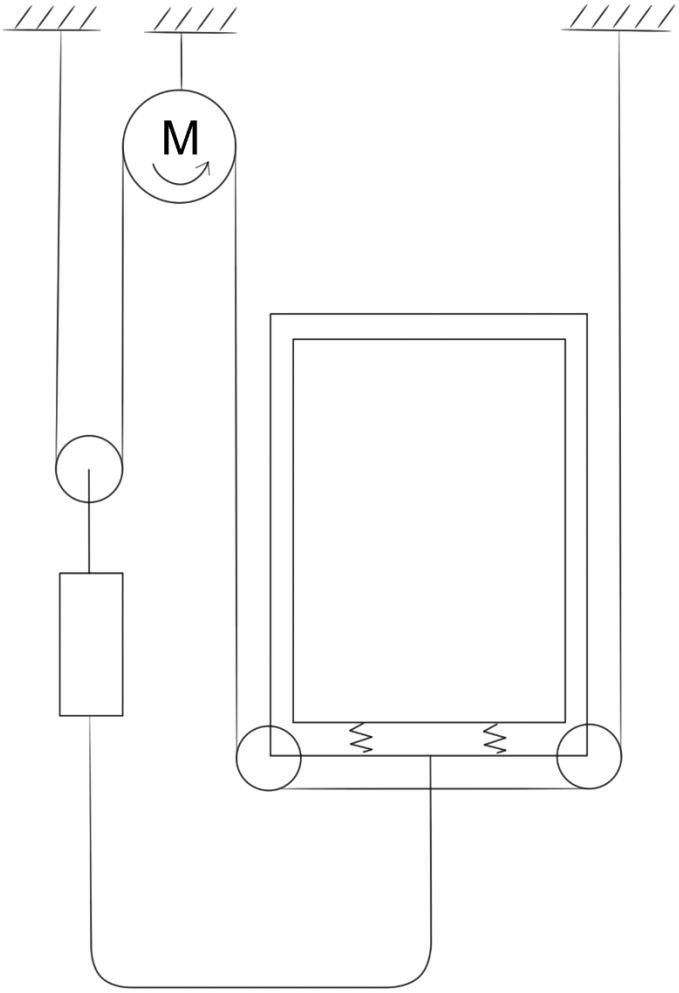


Fig 1 Project pilot elevator

3.2 Parameters

Theoretically the mass of each main component, the stiffness and damping of each connection structure are all required for vibration analysis. However, some mechanical parameters like stiffness and damping can barely be directly measured, while basic parameters like mass and travel route are not provided yet, most of these parameters are estimated or chosen by reference and experience. The value of each parameter in current model are listed in Table 1 with explanation on how they are measured or calculated.

Table 1 Basic elevator parameters

Parameter	**Meaning**	**Value**	**Unit**
Mass	Traction motor	**370**	kg
	Car	1400	
	Car frame	300	
	Counterweight (+frame)	**1035**+150	
Rotary inertia	Traction sheave	40.56	kg-m^2
	Diversion sheave	30	
	Left pulley	25	
	Right pulley	25	
Linear Density	Hoist rope	0.43	kg/m
	Compensation rope	0.43	kg/m
Diameter	Traction sheave	**0.48**	m
	Diversion sheave	**0.4**	
	Pulley	**0.33**	
	Hoist rope	**0.008**	
Stiffness	Traction motor-ground	$6.53*10^6$	N/m
	Spring buffer of rope anchors	$3.75*10^5$	
	Car frame-car (vertical)	$1.8*10^6$	
Damping	*stiffness	0.0001	N/(m/s)
Young modulus	Hoist rope	$9.6*10^7$	Pa
Mass	Sliding guide shoe	13.4	kg
Rotary inertia	Car (x axis)	1728	kg-m^2
	Car frame (x axis)	137	
Stiffness	Guide shoe and car frame	$1.2*10^5$	N/m
	Car frame and car (horizontal)	$2*10^5$	
h1-h8	The location of each force bearing point	-	m

Note: The value in bold are accurate while others are estimated.

3.3 Explanation

To build a mathematical model based on platform like Matlab, the basic structure and mechanical parameters are required. These parameters ought to be decided and recorded during product design, but in this project they are not provided and can only be measured or estimated. The explanations on how to obtain these parameters listed above are given here.

1) Measurement

Basic size information are accessible by regular measurement. The diameter of rope, sheaves and pulleys are measured with tapeline and Vernier caliper.

2) Query and calculation

Some parameters like mass can hardly be directly measured without disassembly, but can be calculated by whole mass distribution. Some parameters can also be given by experience of local maintenance technician.

3) Estimation

For those parameters that are hard for directly measurement or calculated, we can estimate its value according to references. The parameters should be adjusted to best match real world case.

4 DYNAMICS MODELLING

Based on the structure understanding during field visit, the pilot elevator is simplified into a vertical vibration model as well as a horizontal vibration model, in order to study its vibration in vertical and horizontal directions respectively. The kinetic model and equations are thereby built.

4.1 Vertical model

The general form of system dynamics equation can be written as

$$\boldsymbol{M}\ddot{\boldsymbol{X}} + \boldsymbol{C}\dot{\boldsymbol{X}} + \boldsymbol{K}\boldsymbol{X} = \boldsymbol{Q} \tag{1}$$

which consists of the mass matrix $\boldsymbol{M}$, stiffness matrix $\boldsymbol{K}$, damping matrix $\boldsymbol{C}$, load vector $\boldsymbol{Q}$ and the displacement vector $\boldsymbol{X}$. According to the illustration “Mechanical model of elevator vertical vibration” (see page 12) and Lagrange equation, the kinetic equations of elevator vertical vibration is given as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}_i}\right) - \frac{\partial T}{\partial x_i} + \frac{\partial D}{\partial \dot{x}_i} + \frac{\partial U}{\partial x_i} = q_i \tag{2}$$

where x_i, q_i stand for the generalized displacement and generalized force respectively. T, D, U stand for the system kinetic energy, dissipation coefficient and potential energy respectively, which can be given as follows:

$$\begin{aligned} T = {} & \frac{1}{2}m_0\dot{x}_0^2 + \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 + \frac{1}{2}m_3\dot{x}_3^2 + \frac{1}{2}m_4\dot{x}_4^2 + \frac{1}{2}m_5\dot{x}_5^2 + \frac{1}{2}m_6\dot{x}_6^2 + \\ & \frac{1}{2}m_{\mathrm{e}1}\dot{x}_{\mathrm{e}1}^2 + \frac{1}{2}m_{\mathrm{e}2}\dot{x}_{\mathrm{e}2}^2 + \frac{1}{2}m_{\mathrm{e}3}\dot{x}_{\mathrm{e}3}^2 + \frac{1}{2}m_{\mathrm{e}4}\dot{x}_{\mathrm{e}4}^2 + \frac{1}{2}m_{\mathrm{e}5}\dot{x}_{\mathrm{e}5}^2 + \frac{1}{2}m_{\mathrm{e}6}\dot{x}_{\mathrm{e}6}^2 + \\ & \frac{1}{2}J_0\dot{\theta}_0^2 + \frac{1}{2}J_1\dot{\theta}_1^2 + \frac{1}{2}J_2\dot{\theta}_2^2 + \frac{1}{2}J_3\dot{\theta}_3^2 \end{aligned} \tag{3}$$

$$\begin{aligned} D = {} & \frac{1}{2}c_0\dot{x}_0^2 + \frac{1}{2}c_{\mathrm{e}1}\dot{x}_{\mathrm{e}1}^2 + \frac{1}{2}c_{\mathrm{e}2}\dot{x}_{\mathrm{e}4}^2 + \frac{1}{2}c_1(\dot{x}_{\mathrm{e}1} - \dot{x}_1 + r_1\dot{\theta}_1)^2 + \frac{1}{2}c_2(\dot{x}_1 - \dot{x}_{\mathrm{e}2} + r_1\dot{\theta}_1)^2 + \\ & \frac{1}{2}c_2(\dot{x}_{\mathrm{e}2} - \dot{x}_0 + r_0\dot{\theta}_0)^2 + \frac{1}{2}c_3(\dot{x}_0 - \dot{x}_{\mathrm{e}3} + r_0\dot{\theta}_0)^2 + \frac{1}{2}c_3(\dot{x}_{\mathrm{e}3} - \dot{x}_2 + r_2\dot{\theta}_2)^2 + \\ & \frac{1}{2}c_4(\dot{x}_2 - \dot{x}_{\mathrm{e}4} + r_2\dot{\theta}_2)^2 + \frac{1}{2}c_5(\dot{x}_3 - \dot{x}_{\mathrm{e}5} + r_3\dot{\theta}_3)^2 + \frac{1}{2}c_5(\dot{x}_{\mathrm{e}5} - \dot{x}_4)^2 + \\ & \frac{1}{2}c_7(\dot{x}_4 - \dot{x}_1)^2 + \frac{1}{2}c_8(\dot{x}_5 - \dot{x}_2)^2 + \frac{1}{2}c_6(\dot{x}_{\mathrm{e}6} - \dot{x}_5)^2 + \\ & \frac{1}{2}c_6(\dot{x}_3 - \dot{x}_{\mathrm{e}6} + r_3\dot{\theta}_3)^2 + \frac{1}{2}c_9(\dot{x}_5 - \dot{x}_6)^2 + \frac{1}{2}c_{\mathrm{r}}\dot{\theta}_0^2 + \frac{1}{2}c_{10}\dot{x}_3^2 \end{aligned} \tag{4}$$

$$
\begin{aligned}
U = & \frac{1}{2}k_0x_0^2 + \frac{1}{2}k_{\mathrm{e}1}x_{\mathrm{e}1}^2 + \frac{1}{2}k_{\mathrm{e}2}x_{\mathrm{e}4}^2 + \frac{1}{2}k_1(x_{\mathrm{e}1} - x_1 + r_1\theta_1)^2 + \frac{1}{2}k_2(x_1 - x_{\mathrm{e}2} + r_1\theta_1)^2 + \\
& \frac{1}{2}k_2(x_{\mathrm{e}2} - x_0 + r_0\theta_0)^2 + \frac{1}{2}k_3(x_0 - x_{\mathrm{e}3} + r_0\theta_0)^2 + \frac{1}{2}k_3(x_{\mathrm{e}3} - x_2 + r_2\theta_2)^2 + \\
& \frac{1}{2}k_4(x_2 - x_{\mathrm{e}4} + r_2\theta_2)^2 + \frac{1}{2}k_5(x_3 - x_{\mathrm{e}5} + r_3\theta_3)^2 + \frac{1}{2}k_5(x_{\mathrm{e}5} - x_4)^2 + \\
& \frac{1}{2}k_7(x_4 - x_1)^2 + \frac{1}{2}k_8(x_5 - x_2)^2 + \frac{1}{2}k_6(x_{\mathrm{e}6} - x_5)^2 + \\
& \frac{1}{2}k_6(x_3 - x_{\mathrm{e}6} + r_3\theta_3)^2 + \frac{1}{2}k_9(x_5 - x_6)^2 + \frac{1}{2}k_{\mathrm{r}}\theta_0^2 + \frac{1}{2}k_{10}x_3^2
\end{aligned}
\tag{5}
$$

It is worth noticing that the proposed model is of 17 DOF in total, which means the mass, stiffness and damping of each part of hoist rope as well as compensation rope are taken into consideration as well. The tension sheave which doesn't exist in some low speed elevator is also included in the model for the integrality. By considering these potential vibration sources, the proposed model is of generalized ability thus can be applied to various types of elevator with parameter changing.

The symbols *m*, *k*, *c* respectively stands for the mass, stiffness and damping of each part, and *J*, r means the rotary inertia and radius of each sheave or pulley. x and θ indicate the displacement and angular displacement while their derivatives are expressed with an extra sign (point). Detailed corresponding information can be found in the illustration on page 12. Therefore the mass matrix $\boldsymbol{M}$, stiffness matrix $\boldsymbol{K}$, damping matrix $\boldsymbol{C}$ and the displacement vector $\boldsymbol{X}$ in equation (1) are obtained as

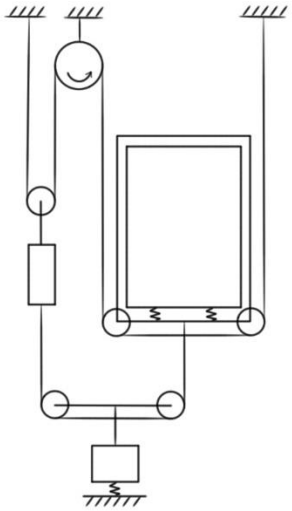
$$\boldsymbol{M} = \mathrm{diag}\left\{m_0, m_1, m_2, m_3, m_4, m_5, m_6, m_{\mathrm{e}1}, m_{\mathrm{e}2}, m_{\mathrm{e}3}, m_{\mathrm{e}4}, m_{\mathrm{e}5}, m_{\mathrm{e}6}, J_0, J_1, J_2, J_3\right\} \quad (6)$$

$$\boldsymbol{K} = \begin{bmatrix}
k_0+k_2+k_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_2 & -k_3 & 0 & 0 & 0 & k_3r_0-k_2r_0 & 0 & 0 & 0 \\
0 & k_1+k_2+k_7 & 0 & 0 & -k_7 & 0 & 0 & -k_1 & -k_2 & 0 & 0 & 0 & 0 & 0 & k_2r_1-k_1r_1 & 0 & 0 \\
0 & 0 & k_3+k_4+k_8 & 0 & 0 & -k_8 & 0 & 0 & 0 & -k_3 & -k_4 & 0 & 0 & 0 & 0 & k_4r_2-k_3r_2 & 0 \\
0 & 0 & 0 & k_5+k_6+k_{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_5 & -k_6 & 0 & 0 & 0 & k_5r_3+k_6r_3 \\
0 & -k_7 & 0 & 0 & k_5+k_7 & 0 & 0 & 0 & 0 & 0 & 0 & -k_5 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -k_8 & 0 & 0 & k_6+k_8+k_9 & -k_9 & 0 & 0 & 0 & 0 & 0 & -k_6 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -k_9 & k_9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -k_1 & 0 & 0 & 0 & 0 & 0 & k_{e1}+k_1 & 0 & 0 & 0 & 0 & 0 & 0 & k_1r_1 & 0 & 0 \\
-k_2 & -k_2 & 0 & 0 & 0 & 0 & 0 & 0 & 2k_2 & 0 & 0 & 0 & 0 & k_2r_0 & -k_2r_1 & 0 & 0 \\
-k_3 & 0 & -k_3 & 0 & 0 & 0 & 0 & 0 & 0 & 2k_3 & 0 & 0 & 0 & -k_3r_0 & 0 & k_3r_2 & 0 \\
0 & 0 & -k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_{e2}+k_4 & 0 & 0 & 0 & 0 & -k_4r_2 & 0 \\
0 & 0 & 0 & -k_5 & -k_5 & 0 & 0 & 0 & 0 & 0 & 0 & 2k_5 & 0 & 0 & 0 & 0 & -k_5r_3 \\
0 & 0 & 0 & -k_6 & 0 & -k_6 & 0 & 0 & 0 & 0 & 0 & 0 & 2k_6 & 0 & 0 & 0 & -k_6r_3 \\
k_3r_0-k_2r_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_2r_0 & -k_3r_0 & 0 & 0 & 0 & k_{\mathrm{r}}+(k_2+k_3)r_0^2 & 0 & 0 & 0 \\
0 & k_2r_1-k_1r_1 & 0 & 0 & 0 & 0 & 0 & k_1r_1 & -k_2r_1 & 0 & 0 & 0 & 0 & 0 & (k_1+k_2)r_1^2 & 0 & 0 \\
0 & 0 & k_4r_2-k_3r_2 & 0 & 0 & 0 & 0 & 0 & 0 & k_3r_2 & -k_4r_2 & 0 & 0 & 0 & 0 & (k_3+k_4)r_2^2 & 0 \\
0 & 0 & 0 & k_5r_3+k_6r_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_5r_3 & -k_6r & 0 & 0 & 0 & (k_5+k_6)r_3^2
\end{bmatrix} \quad (7)$$

$$\boldsymbol{C} = \begin{bmatrix}
c_0+c_2+c_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_2 & -c_3 & 0 & 0 & 0 & c_3r_0-c_2r_0 & 0 & 0 & 0 \\
0 & c_1+c_2+c_7 & 0 & 0 & -c_7 & 0 & 0 & -c_1 & -c_2 & 0 & 0 & 0 & 0 & 0 & c_2r_1-c_1r_1 & 0 & 0 \\
0 & 0 & c_3+c_4+c_8 & 0 & 0 & -c_8 & 0 & 0 & 0 & -c_3 & -c_4 & 0 & 0 & 0 & 0 & c_4r_2-c_3r_2 & 0 \\
0 & 0 & 0 & c_5+c_6+c_{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_5 & -c_6 & 0 & 0 & 0 & c_5r_3+c_6r_3 \\
0 & -c_7 & 0 & 0 & c_5+c_7 & 0 & 0 & 0 & 0 & 0 & 0 & -c_5 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -c_8 & 0 & 0 & c_6+c_8+c_9 & -c_9 & 0 & 0 & 0 & 0 & 0 & -c_6 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -c_9 & c_9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -c_1 & 0 & 0 & 0 & 0 & 0 & c_{e1}+c_1 & 0 & 0 & 0 & 0 & 0 & 0 & c_1r_1 & 0 & 0 \\
-c_2 & -c_2 & 0 & 0 & 0 & 0 & 0 & 0 & 2c_2 & 0 & 0 & 0 & 0 & c_2r_0 & -c_2r_1 & 0 & 0 \\
-c_3 & 0 & -c_3 & 0 & 0 & 0 & 0 & 0 & 0 & 2c_3 & 0 & 0 & 0 & -c_3r_0 & 0 & c_3r_2 & 0 \\
0 & 0 & -c_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_{e2}+c_4 & 0 & 0 & 0 & 0 & -c_4r_2 & 0 \\
0 & 0 & 0 & -c_5 & -c_5 & 0 & 0 & 0 & 0 & 0 & 0 & 2c_5 & 0 & 0 & 0 & 0 & -c_5r_3 \\
0 & 0 & 0 & -c_6 & 0 & -c_6 & 0 & 0 & 0 & 0 & 0 & 0 & 2c_6 & 0 & 0 & 0 & -c_6r_3 \\
c_3r_0-c_2r_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_2r_0 & -c_3r_0 & 0 & 0 & 0 & c_{\mathrm{r}}+(c_2+c_3)r_0^2 & 0 & 0 & 0 \\
0 & c_2r_1-c_1r_1 & 0 & 0 & 0 & 0 & 0 & c_1r_1 & -c_2r_1 & 0 & 0 & 0 & 0 & 0 & (c_1+c_2)r_1^2 & 0 & 0 \\
0 & 0 & c_4r_2-c_3r_2 & 0 & 0 & 0 & 0 & 0 & 0 & c_3r_2 & -c_4r_2 & 0 & 0 & 0 & 0 & (c_3+c_4)r_2^2 & 0 \\
0 & 0 & 0 & c_5r_3+c_6r_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_5r_3 & -c_6r & 0 & 0 & 0 & (c_5+c_6)r_3^2
\end{bmatrix} \quad (8)$$

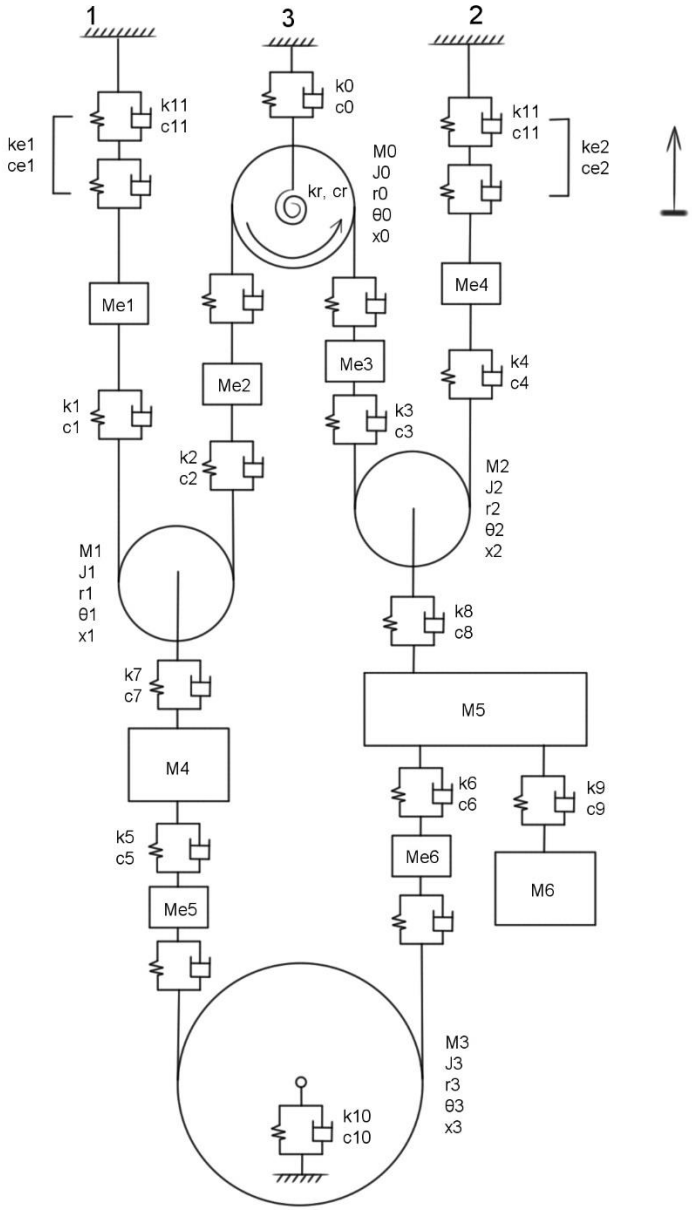
$$\boldsymbol{X} = \left\{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_{\mathrm{e}1}, x_{\mathrm{e}2}, x_{\mathrm{e}3}, x_{\mathrm{e}4}, x_{\mathrm{e}5}, x_{\mathrm{e}6}, \theta_0, \theta_1, \theta_2, \theta_3\right\}^{\mathrm{T}} \quad (9)$$

MECHANICAL MODEL OF ELEVATOR VERTICAL VIBRATION



2:1 type

17 DOF



M0, J0, r0, θ0, x0: The mass, rotary inertia, radius, rotation angle and displacement of traction sheave.

M1, J1, r1, θ1, x1: The mass, rotary inertia, radius, rotation angle and displacement of rope sheave (for counterweight).

M2, J2, r2, θ2, x2: The mass, rotary inertia, radius, rotation angle and displacement of rope sheave (for car).

M3, J3, r3, θ3, x3: The mass, rotary inertia, radius, rotation angle and displacement of tension pulley.

M4, M5, M6: Mass of counterweight, lifting frame and car & inner load.

Me1. Me2, Me3, Me4, Me5, Me6: The equivalent mass of corresponding part of hoist rope.

ke1, ce1, ke2, ce2: The equivalent stiffness and damping of the hoist rope and connected spring at both ends.

kr, cr: The rotational stiffness and damping of traction sheave

k(n), c(n): The stiffness and damping of each individual part.

x(n): The vertical displacement of each induvidual mass.

DRAFT

4.2 Horizontal model

According to Newton's second law, the differential equations of car horizontal vibration is available and can be written as

$$\boldsymbol{M}\ddot{\boldsymbol{X}} + \boldsymbol{C}\dot{\boldsymbol{X}} + \boldsymbol{K}\boldsymbol{X} = \boldsymbol{Q} \tag{10}$$

Likewise, the symbols m, J, k and c respectively stands for the mass, rotary inertia, stiffness and damping of each part in horizontal direction. x and θ indicate the displacement and angular displacement while their derivatives are expressed with an extra sign (point). h, l are length symbols and F indicates external force. Detailed corresponding information of these symbols can also be found in illustration on page 14.

Then the mass matrix $\boldsymbol{M}$, stiffness matrix $\boldsymbol{K}$, damping matrix $\boldsymbol{C}$, stimulation vector $\boldsymbol{Q}$ and the vector $\boldsymbol{X}$ are cast as

$$\boldsymbol{M} = \mathrm{diag}\left\{m_0, m_\mathrm{f}, m_1, m_2, m_3, m_4, J_0, J_\mathrm{f}\right\} \tag{11}$$

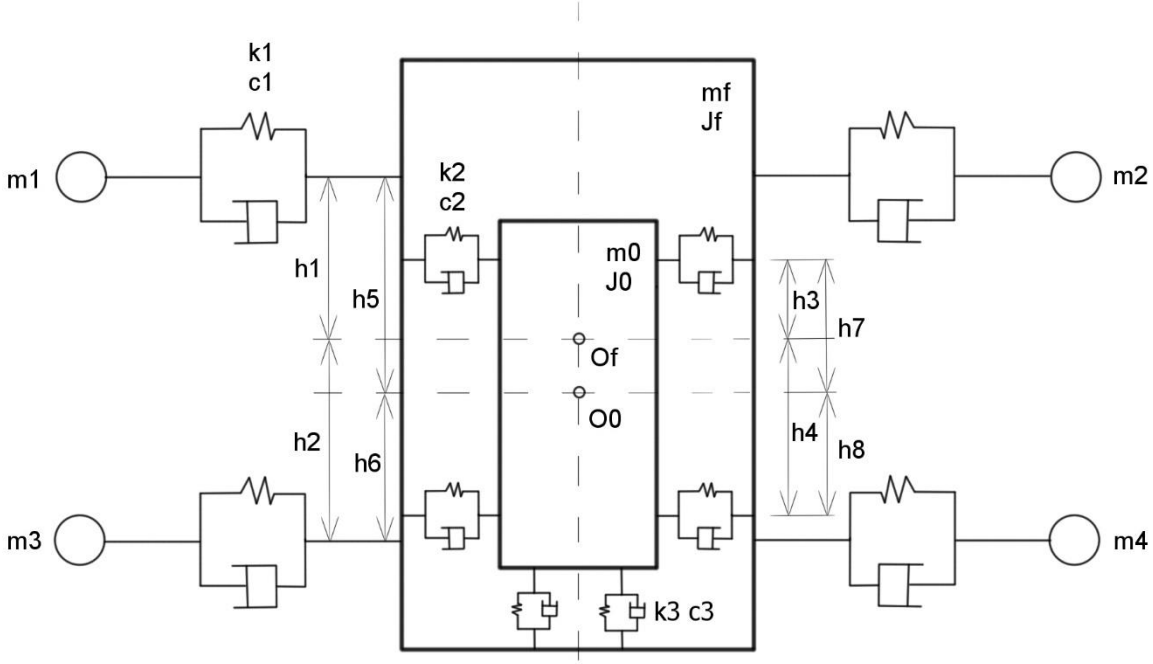
$$\boldsymbol{K} = \begin{bmatrix} -4k_2 & 4k_2 & 0 & 0 & 0 & 0 & 2k_2h_8 - 2k_2h_7 & 2k_2h_3 - 2k_2h_4 \\ 4k_2 & -4k_1 - 4k_2 & k_1 & k_1 & k_1 & k_1 & 2k_2h_7 - 2k_2h_8 & -2k_1h_1 + 2k_1h_2 - 2k_2h_3 + 2k_2h_4 \\ 0 & k_1 & -k_1 & 0 & 0 & 0 & 0 & k_1h_1 \\ 0 & k_1 & 0 & -k_1 & 0 & 0 & 0 & k_1h_1 \\ 0 & k_1 & 0 & 0 & -k_1 & 0 & 0 & -k_1h_2 \\ 0 & k_1 & 0 & 0 & 0 & -k_1 & 0 & -k_1h_2 \\ 2k_2h_8 - 2k_2h_7 & 2k_2h_7 - 2k_2h_8 & 0 & 0 & 0 & 0 & -2k_2h_7^2 - 2k_2h_8^2 - 2k_2l^2 & 2k_2h_3h_7 + 2k_2h_4h_8 + 2k_3l^2 \\ 2k_2h_3 - 2k_2h_4 & -2k_1h_1 + 2k_1h_2 - 2k_2h_3 + 2k_2h_4 & k_1h_1 & k_1h_1 & -k_1h_2 & -k_1h_2 & 2k_2h_3h_7 + 2k_2h_4h_8 + 2k_3l^2 & -2k_1h_1^2 - 2k_1h_2^2 - 2k_2h_3^2 - 2k_2h_8^2 - 2k_3l^2 \end{bmatrix} \tag{12}$$

$$\boldsymbol{C} = \begin{bmatrix} -4c_2 & 4c_2 & 0 & 0 & 0 & 0 & 2c_2h_8 - 2c_2h_7 & 2c_2h_3 - 2c_2h_4 \\ 4c_2 & -4c_1 - 4c_2 & c_1 & c_1 & c_1 & c_1 & 2c_2h_7 - 2c_2h_8 & -2c_1h_1 + 2c_1h_2 - 2c_2h_3 + 2c_2h_4 \\ 0 & c_1 & -c_1 & 0 & 0 & 0 & 0 & c_1h_1 \\ 0 & c_1 & 0 & -c_1 & 0 & 0 & 0 & c_1h_1 \\ 0 & c_1 & 0 & 0 & -c_1 & 0 & 0 & -c_1h_2 \\ 0 & c_1 & 0 & 0 & 0 & -c_1 & 0 & -c_1h_2 \\ 2c_2h_8 - 2c_2h_7 & 2c_2h_7 - 2c_2h_8 & 0 & 0 & 0 & 0 & -2c_2h_7^2 - 2c_2h_8^2 - 2c_2l^2 & 2c_2h_3h_7 + 2c_2h_4h_8 + 2c_3l^2 \\ 2c_2h_3 - 2c_2h_4 & -2c_1h_1 + 2c_1h_2 - 2c_2h_3 + 2c_2h_4 & c_1h_1 & c_1h_1 & -c_1h_2 & -c_1h_2 & 2c_2h_3h_7 + 2c_2h_4h_8 + 2c_3l^2 & -2c_1h_1^2 - 2c_1h_2^2 - 2c_2h_3^2 - 2c_2h_8^2 - 2c_3l^2 \end{bmatrix} \tag{13}$$

$$\boldsymbol{X} = \left\{x_0, x_\mathrm{f}, x_1, x_2, x_3, x_4, \theta_0, \theta_\mathrm{f}\right\}^\mathrm{T} \tag{14}$$

$$\boldsymbol{Q} = \left\{0, \sum_{i=1}^{4} F_i, F_1, F_2, F_3, F_4, 0, (\sum_{i=1}^{2} F_i h_1 + \sum_{i=3}^{4} F h_2)\right\}^\mathrm{T} \tag{15}$$

MECHANICAL MODEL OF ELEVATOR HORIZONTAL VIBRATION



8 DOF

m0, J0: The mass and rotary inertia of the lift car.

mf, Jf: The mass and rotatry inertia of car frame.

O0, Of: The center of mass of the lift car and car frame.

m1, m2, m3, m4: The mass of each guide shoe.

k1, c1, k2, c2, k3, c3: The stiffness and damping of 1) each guide shoe;
2) horizontal damper between lift car and car frame and
3) vertical damper between lift car and car frame, respectively.

h1, h2, h3, h4, h5, h6, h7, h8: The distance between each point of action and the center of mass, as labeled.

DRAFT

5 EQUATION SOLUTION

According to aforementioned system of ordinary differential equations based on proposed model, the solution of vertical vibration is of 17 vibration vectors corresponding to each component and 8 vibration vectors for horizontal case. The proposed model is of complexity due to multiple components, varying car speed and varying mass/stiffness distribution, thus its analytical solution is hardly available. In this section two numerical solution methodologies (ODE45 and Newmark-beta method) are introduced combined with their use on Matlab platform. ODE is actually a series of solving functions in Matlab, which aims at solving ordinary differential equations, we use ode45 solver in this case for its accuracy and low computation cost.

5.1 ODE

ODE is a series of solution functions in Matlab specific for solving system of ordinary differential equations. Table 2 lists the initial value problem solvers, the kind of problem each solver are specified for, and the method each solver uses.

Table 2 ODE solvers

Solver	Solves These Kinds of Problems	Method
ode45	Nonstiff differential equations	Runge-Kutta
ode23	Nonstiff differential equations	Runge-Kutta
ode113	Nonstiff differential equations	Adams
ode15s	Stiff differential equations and DAEs	NDFs (BDFs)
ode23s	Stiff differential equations	Rosenbrock
ode23t	Moderately stiff differential equations and DAEs	Trapezoidal rule
ode23tb	Stiff differential equations	TR-BDF2
ode15i	Fully implicit differential equations	BDFs

ODE45 is based on an explicit Runge-Kutta [15-16] formula, the Dormand-Prince pair. It is a one-step solver in computing $y(t_n)$, it needs only the solution at the immediately preceding time point, $y(t_{n-1})$. In general, ode45 is the best function to apply as a first try for most problems. [17] The content of explicit Runge-Kutta method can be briefly introduced as follow:

The family of explicit Runge–Kutta methods is given by

$$y_{n+1} = y_n + \sum_{i=1}^{s} h b_i k_i \tag{16}$$

where

$$\begin{aligned} k_1 &= f(t_n, y_n) \\ k_2 &= f(t_n + c_2 h, y_n + h(a_{21} k_1)) \\ k_3 &= f(t_n + c_3 h, y_n + h(a_{31} k_1 + a_{32} k_2)) \\ &\cdots \\ k_s &= f(t_n + c_s h, y_n + h(a_{s1} k_1 + a_{s2} k_2 + \cdots + a_{s,s-1} k_{s-1})) \end{aligned} \tag{17}$$

To specify a particular method, one needs to provide the integer s (the number of stages), and the coefficients a_{ij} (for $1 \le j < i \le s$), b_i (for i = 1, 2, ..., s) and c_i (for i = 2, 3, ..., s). The matrix $[a_{ij}]$ is called the *Runge–Kutta matrix*, while the b_i and c_i are known as the *weights* and the *nodes*.

$$\begin{array}{c|cccc} 0 & & & & \\ c_2 & a_{21} & & & \\ c_3 & a_{31} & a_{32} & & \\ \vdots & \vdots & & \ddots & \\ c_s & a_{s1} & a_{s2} & \cdots & a_{s,s-1} \\ \hline & b_1 & b_2 & \cdots b_{s-1} & b_s \end{array}$$

The Runge–Kutta method is consistent if

$$\sum_{j=1}^{i-1} a_{ij} = c_i \text{ for } i = 2, \ldots, s. \tag{18}$$

There are also accompanying requirements if one requires the method to have a certain order p, meaning that the local truncation error is $O(h^{p+1})$. These can be derived from the definition of the truncation error itself. For example, a two-stage method has order 2 if $b_1 + b_2 = 1$, $b_2 c_2 = 1/2$, and $a_{21} = c_2$.

Here the ODE45 solver on Matlab platform is actually used to solve the proposed vertical and horizontal vibration models. Matlab ODE solvers accept only first-order differential equations. To use the solvers with higher-order ODEs, we first rewrite each equation as an equivalent system of first-order differential equations of the form. The achieved solution in both free vibration and forced vibration case will be presented in section 6 and 7.

5.2 Newmark- β

The Newmark-beta method is a method of numerical integration used to solve differential equations. It is widely used in numerical evaluation of the dynamic response of structures and solids such as in finite element analysis to model dynamic systems. The method is named after Nathan M. Newmark, former Professor of Civil Engineering at the University of Illinois, who developed it in 1959 for use in Structural dynamics. Its key algorithm can be given as

$$\{\dot{u}\}_{t+\Delta t} = \{\dot{u}\}_t + [(1-\beta)\{\ddot{u}\}_t + \beta\{\ddot{u}\}_{t+\Delta t}]\Delta t \quad (19)$$

$$\{u\}_{t+\Delta t} = \{u\}_t + \{\dot{u}\}\Delta t + [(\frac{1}{2}-\gamma)\{\ddot{u}\}_t + \gamma\{\ddot{u}\}_{t+\Delta t}]\Delta t^2 \quad (20)$$

where $\{u\}_t$、$\{\dot{u}\}_t$、$\{\ddot{u}\}_t$ stands for the displacement, velocity and acceleration respectively. β=0.5, γ=0.25. The procedures can be summarized as follows:

1) Construct the mass matrix **M**, stiffness matrix **K** and damping matrix **C**;

2) Decide the initial values $\{u\}_0$, $\{\dot{u}\}_0$ and $\{\ddot{u}\}_0$;

3) Choose the integration step Δt and calculate the integration constant

$$\alpha_0 = \frac{1}{\gamma\Delta t^2},\ \alpha_1 = \frac{\beta}{\gamma\Delta t},\ \alpha_2 = \frac{1}{\gamma\Delta t},\ \alpha_3 = \frac{1}{2\gamma}-1,$$

$$\alpha_4 = \frac{\beta}{\gamma}-1,\ \alpha_5 = \frac{\Delta t}{2}(\frac{\beta}{\gamma}-2),\ \alpha_6 = \Delta t(1-\beta),\ \alpha_7 = \beta\Delta t;$$

4) Construct the effective stiffness matrix $[\bar{K}] = [K] + \alpha_0[M] + \alpha_1[C]$;

5) Calculate the velocity and acceleration at time $t + \Delta t$

$$\{\ddot{u}\}_{t+\Delta t} = \alpha_0(\{u\}_{t+\Delta t} - \{u\}_t) - \alpha_2\{\dot{u}\}_t - \alpha_3\{\ddot{u}\}_t$$

$$\{\dot{u}\}_{t+\Delta t} = \{\dot{u}\}_t + \alpha_6\{\ddot{u}\}_t + \alpha_7\{\ddot{u}\}_{t+\Delta t}$$

6 FREE VIBRATION ANALYSIS

The elevator vibration response consists of two parts, i.e., free vibration and forced vibration. The former result from initial conditions, with characteristic frequencies equal to the natural frequencies. In this section each order of natural frequencies in both vertical and horizontal direction are predicted using proposed vibration model. Notice that the characteristic frequencies are time-varying due to varying rope mass and stiffness. The time domain, frequency domain and joint time-frequency analysis are used.

6.1 Natural frequencies

6.1.1 Vertical natural frequencies

The natural frequencies of each order can be cast as the eigenvalue of the matrix $\boldsymbol{M}^{-1}\boldsymbol{K}$. Actually the 17 order of natural frequencies are all time varying, and can be recorded as 17 frequency vectors. To provide a better view for identification, only three points (beginning, middle and end) are plotted to represent the change. Given the required mass matrix and stiffness matrix at these three certain moment, the natural frequencies can be calculated and plotted in **Figure 2** as

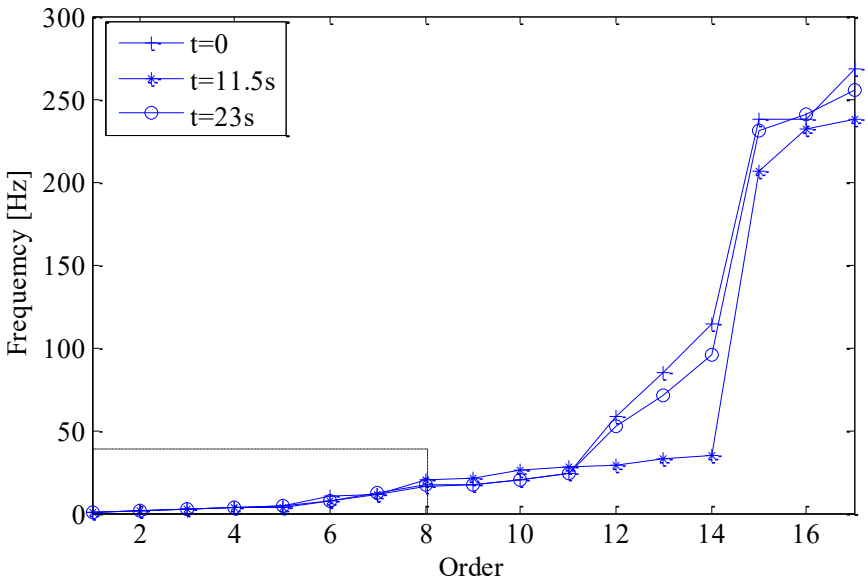


Fig 2 Vertical natural frequencies

It is obvious that the range of natural frequencies are limited, especially in low frequency range which are close to the characteristic frequencies range which can be perceived by human being (as highlighted in the figure by dash rectangle). The detailed time varying feature of low order natural frequencies are exhibited in **Figure 3**.

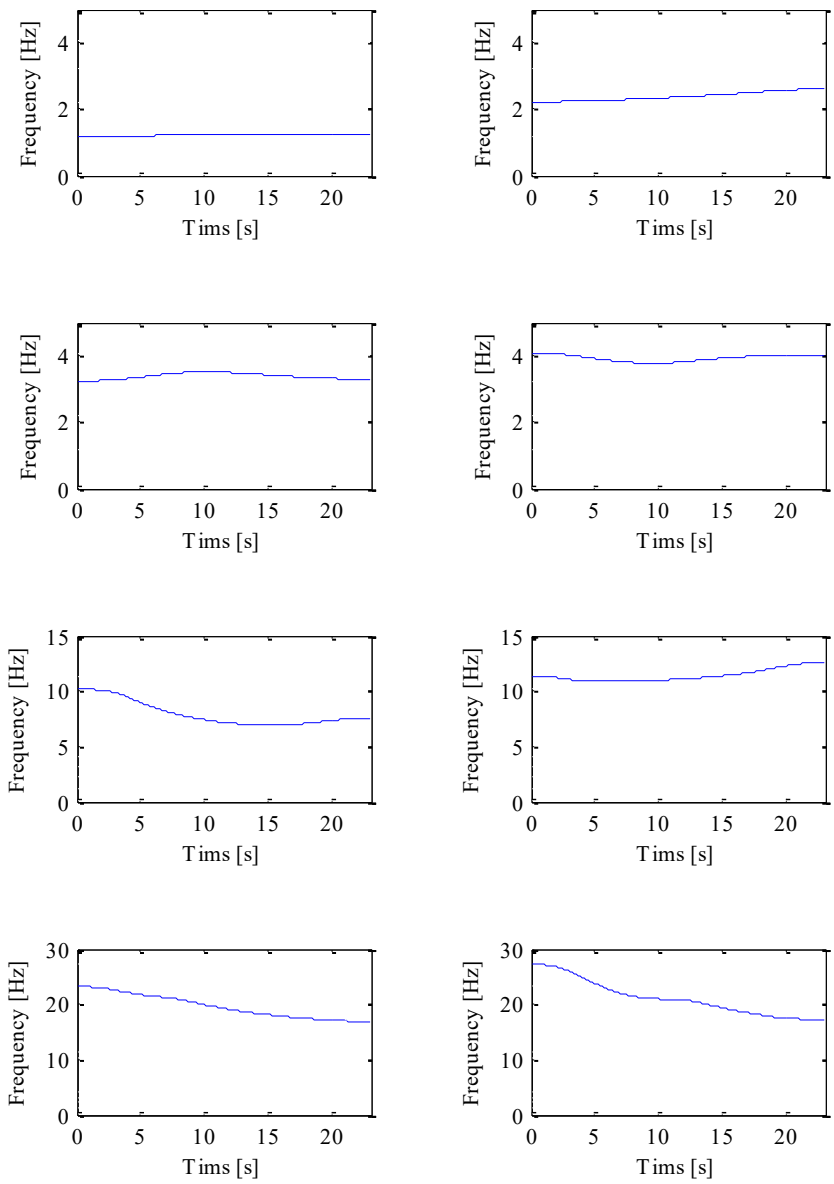


Fig 3 Vertical natural frequencies (low order)

6.1.2 Horizontal natural frequencies

The horizontal natural frequencies are constant. Likewise, the natural frequencies of each order can be cast as the eigenvalue of the matrix $\boldsymbol{M}^{-1}\boldsymbol{K}$. Given the required mass matrix and stiffness matrix, the natural frequencies can be calculated and plotted in **Figure 4** as

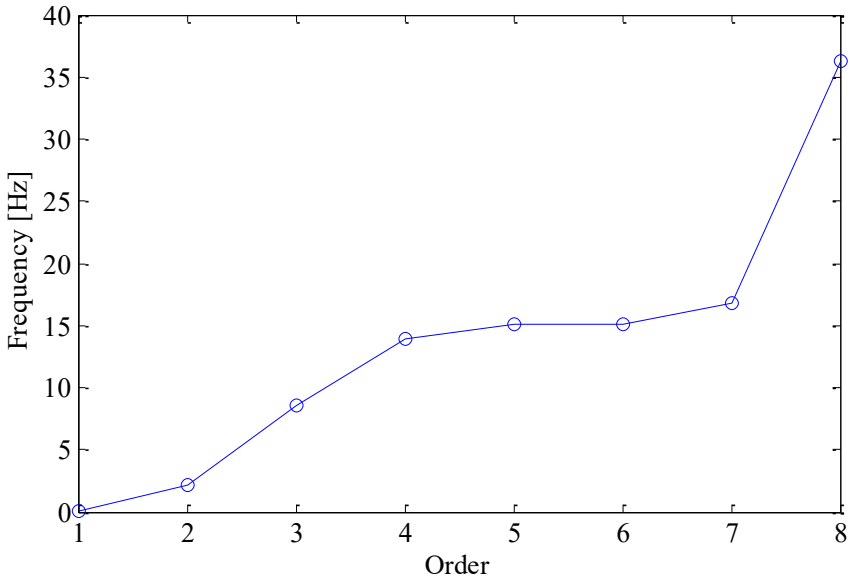
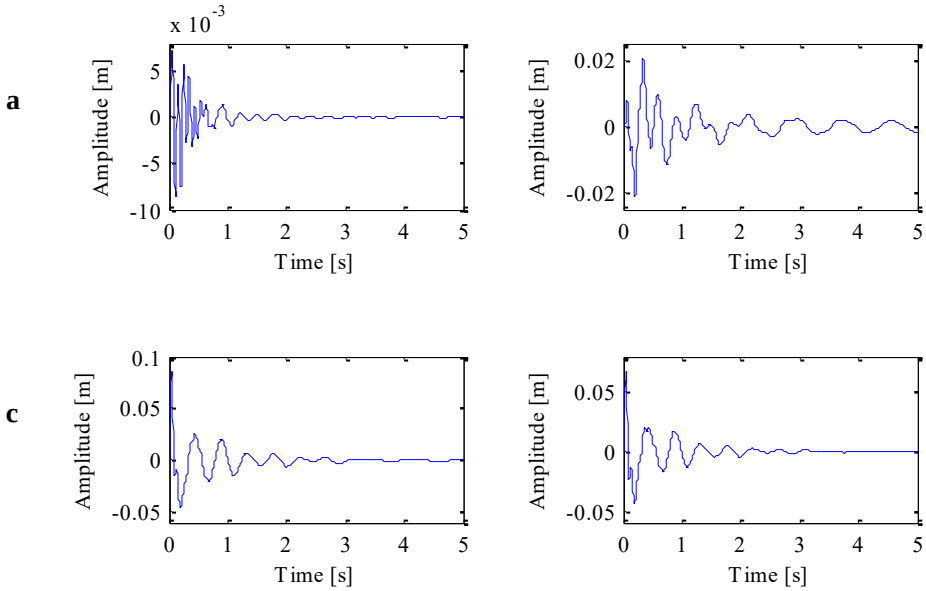


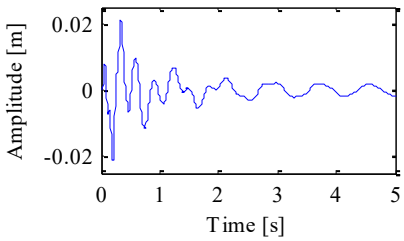
Fig 4 Horizontal natural frequencies

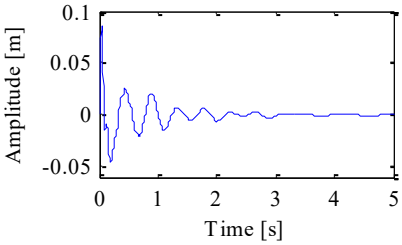
6.2 Vertical vibration analysis

Given the lift car initial vertical displacement of 0.1m, while other initial values are all set to be 0, the vibration response of each order can be obtained as presented in **Figure 5 (a-q)**.



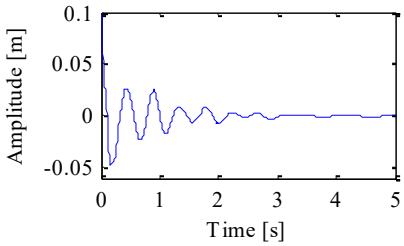
e

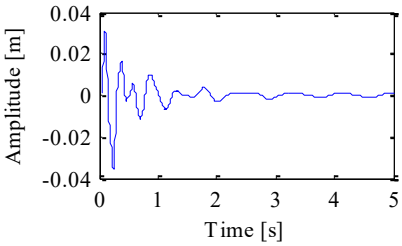

Amplitude [m]
0.02
0
-0.02
0 1 2 3 4 5
Time [s]


Amplitude [m]
0.1
0.05
0
-0.05
0 1 2 3 4 5
Time [s]

f

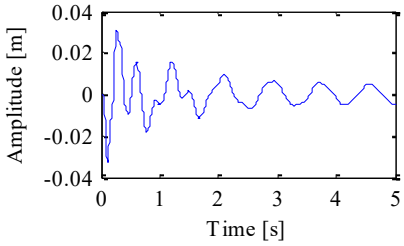
g

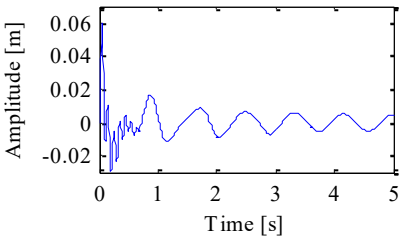

Amplitude [m]
0.1
0.05
0
-0.05
0 1 2 3 4 5
Time [s]


Amplitude [m]
0.04
0.02
0
-0.02
-0.04
0 1 2 3 4 5
Time [s]

h

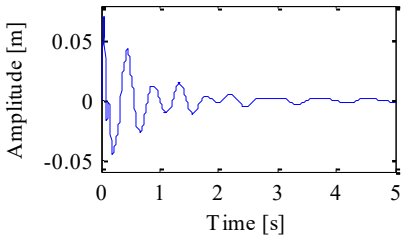
i

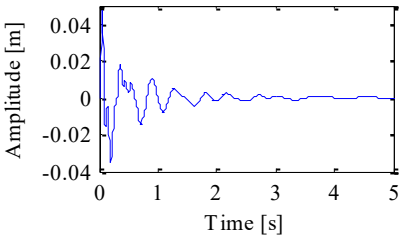

Amplitude [m]
0.04
0.02
0
-0.02
-0.04
0 1 2 3 4 5
Time [s]


Amplitude [m]
0.06
0.04
0.02
0
-0.02
0 1 2 3 4 5
Time [s]

j

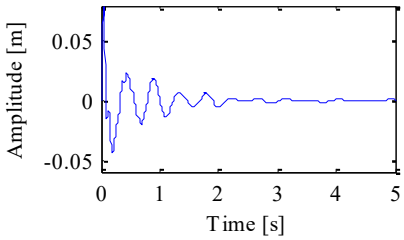
k

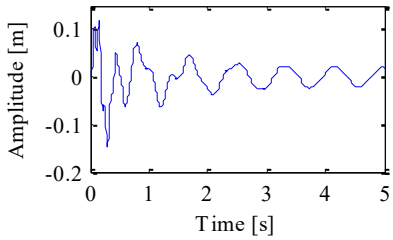

Amplitude [m]
0.05
0
-0.05
0 1 2 3 4 5
Time [s]


Amplitude [m]
0.04
0.02
0
-0.02
-0.04
0 1 2 3 4 5
Time [s]

l

m


Amplitude [m]
0.05
0
-0.05
0 1 2 3 4 5
Time [s]


Amplitude [m]
0.1
0
-0.1
-0.2
0 1 2 3 4 5
Time [s]

n

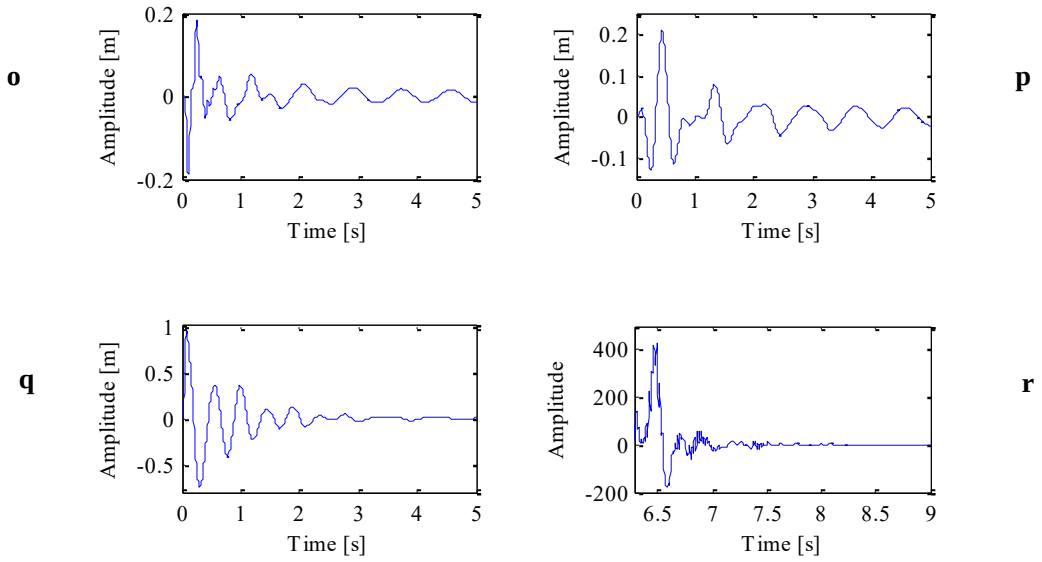
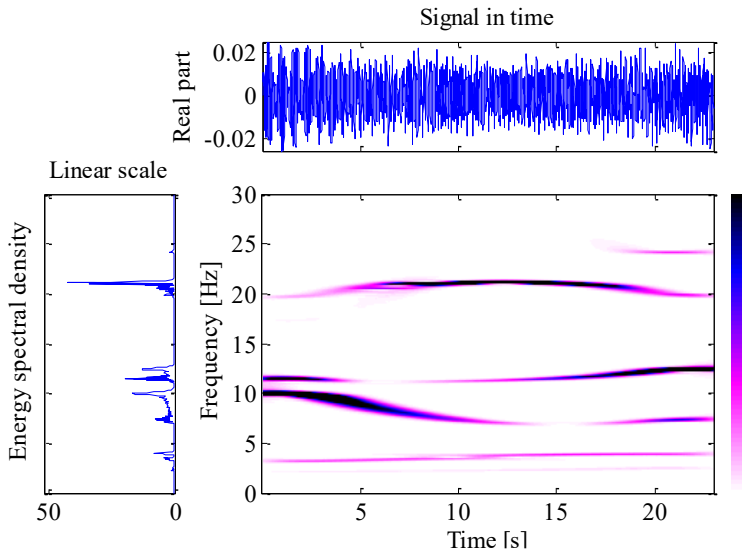


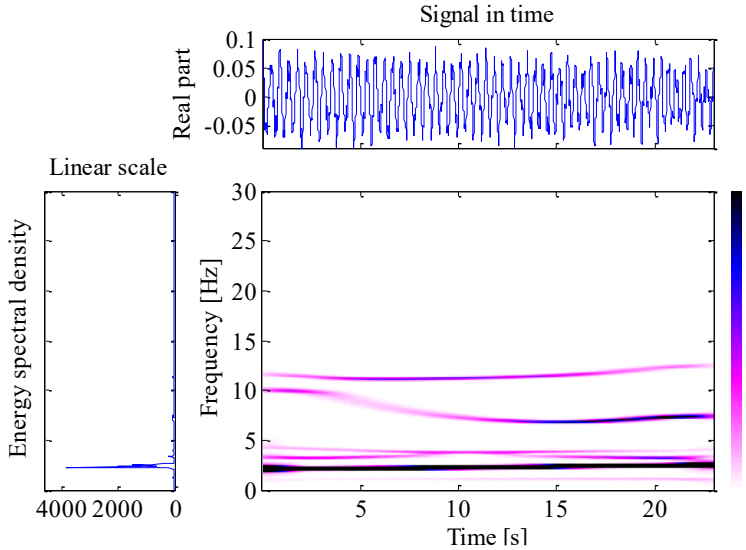
Fig 6 Vertical free vibration

Each order of vertical free vibration has a similar waveform structure, i.e., quick vibration attenuation within 2s. This feature is consistent with the in-situ sensor perceived car vertical vibration during brake process, as shown in **Figure 6(r)**.

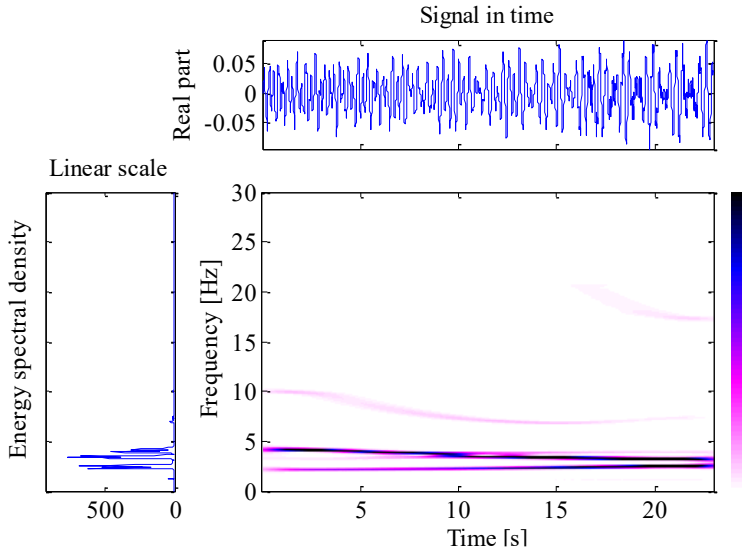
To better understand the characteristic frequencies constitution in free vibration, we set the damping=0 to avoid energy sharply attenuation. The time frequency representation of main parts (motor, car and both rope ends) vibration are calculated by short time Fourier transform (STFT) and exhibited in **Figure 7**.



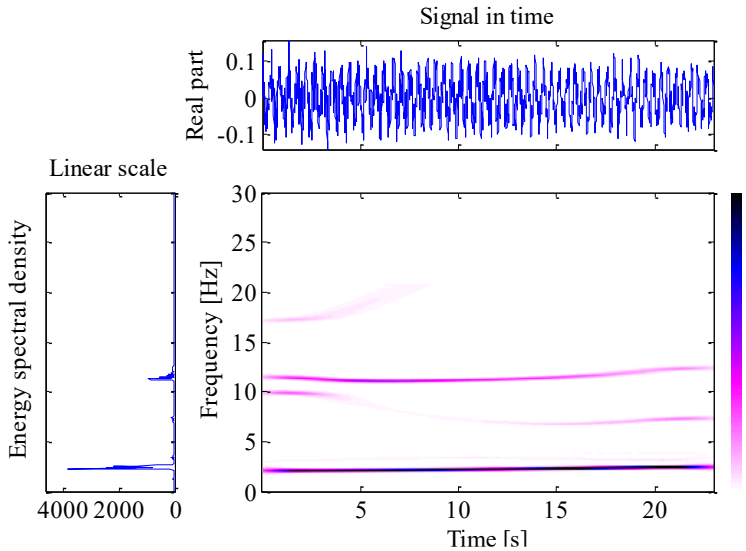
(a) TFR of motor vibration signal (free vibration)



(b) TFR of car vibration signal (free vibration)



(c) TFR of rope end (counterweight side) vibration signal (free vibration)



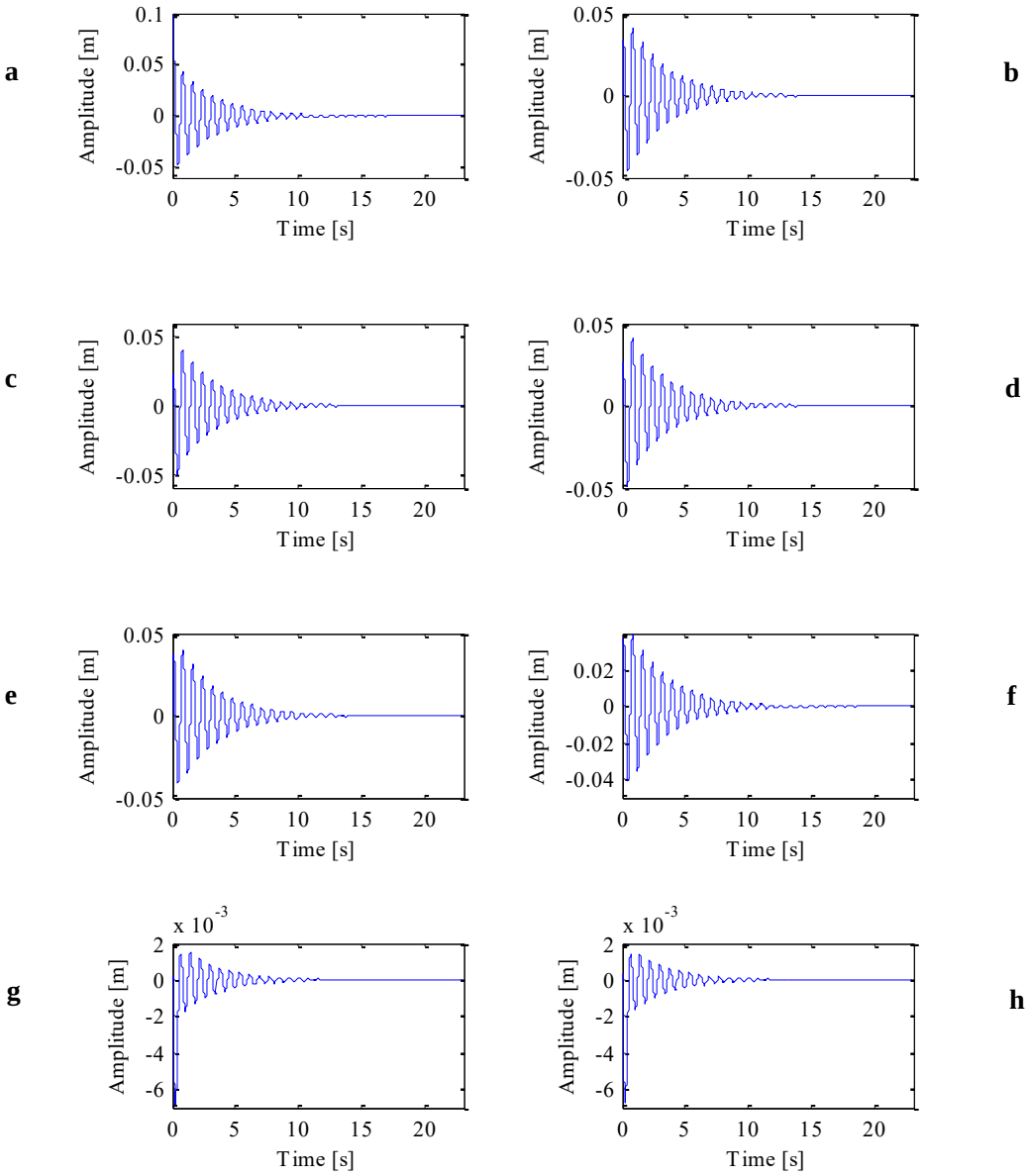
(d) TFR of rope end (car side) vibration signal (free vibration)

Fig 7 TFR of vertical free vibration

From the TFRs of vertical vibration signals, we can clearly identify several dominant characteristic frequency components, which are time-varying instantaneous frequency. They are validated as each order of vertical natural frequencies, thus the frequency component constitution of free vibration signals is consistent with natural frequencies distribution shown in **Fig 3.**

6.3 Horizontal vibration analysis

Givee the lift car initial horizontal displacement of 0.1m, while other initial values are all set to be 0, the vibration response of each order can be obtained as presented in **Figure 8 (a-h)**.



i

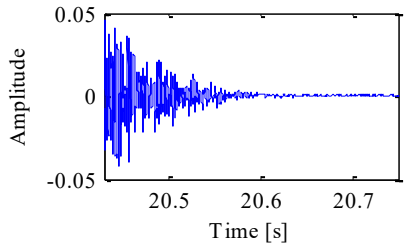


Fig 8 Horizontal free vibration

Comparing the model calculated car horizontal vibration waveform with that in real world case, similar attenuation feature are also visible. The Fourier spectrum analysis is used here for frequency constitution revealing since the characteristic frequencies are constant. Set all the damping to be zero, the Fourier spectrum of car horizontal vibration signal is achieved, as plotted in Figure 9.

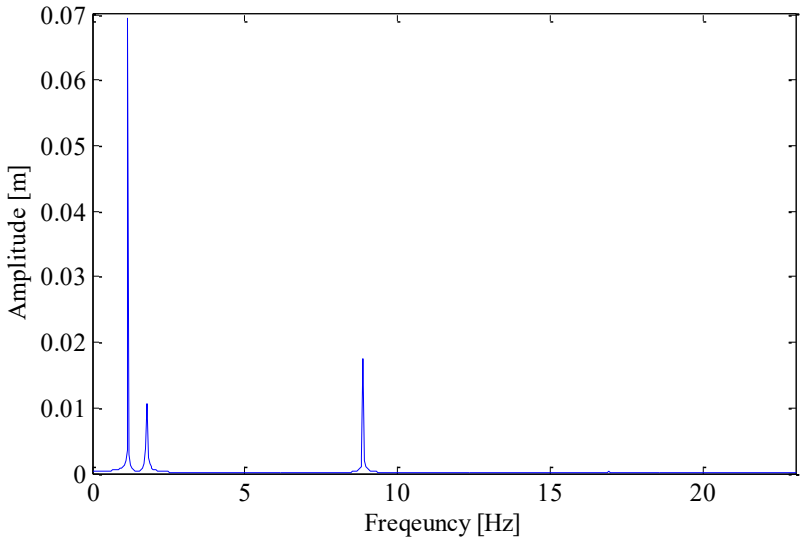


Fig 9 Fourier spectrum of car horizontal free vibration

The three peaks in **Figure 9** respectively stands for the 2nd order of vertical natural frequency and the 2nd as well as 3rd order of horizontal natural frequencies as given in Figure 3-4, therefore the horizontal inherent feature can also be extracted by vibration signal.

6.4 Conclusion

In analysis above, the natural frequencies in both vertical and horizontal direction have been obtained. Given an initial car displacement we mimic the passenger entering or emergency brake process, in which free vibration occurs. The time-frequency analysis of each order of free vibration signals indicate that natural frequencies are the main frequency components in free vibration, and traction motor as well as rope end (counterweight side) can provide better frequency characteristic in vertical direction. Besides, the analysis of horizontal free vibration shows that low order vertical natural frequency component may also exist in horizontal vibration signals due to the rotary effect of car and/or car frame. These findings will help guide the analysis of actual elevator vibration signals.

Free vibration is an inevitable vibration form which can be useful in complex system vibration monitoring and fault diagnosis. By using an accurate vibration model in which the mass matrix and stiffness matrix are available, researchers and engineers can obtain each order of natural frequencies, which can be viewed as inherent characteristic frequencies in real world analysis. The proposed vertical and horizontal vibration model are capable of predicting natural frequencies and are validated by free vibration signal analysis in both frequency domain and joint time and frequency domain.

7 FORCED VIBRATION ANALYSIS

The forced vibration is actually generated by the driving part in elevator, which may show different features due to variant conditions. Besides, the fault is also a kind of potential excitation. By analyzing the elevator forced vibration in both vertical and horizontal directions, we can get information of elevator running conditions and identify potential fault. Note that the external force or torque may also vary with respect to elevator car position, or the unevenness of guide rail.

7.1 External excitation

7.1.1 Vertical excitation

The vertical vibration of elevator system mainly result from the force set by motor controlling system. Therefore a whole cycle of speed up/down process is in analysis, the set lift car vertical speed is

$$v(t)=\begin{cases}\frac{v_0}{2}(1-\cos\frac{\pi}{t_1}t) & 0\le t\le t_1\\ v_0 & t_1\le t\le t_2\\ \frac{v_0}{2}(1-\cos(\frac{\pi(t_3-t)}{t_3-t_2})) & t_2\le t\le t_3\end{cases} \tag{19}$$

where v_0 stands for the nominal velocity, and $0-t_1$, t_1-t_2 and t_2-t_3 correspond to the three stages of a whole cycle. Given simulated arguments, the car speed is drawn in **Figure 10**.

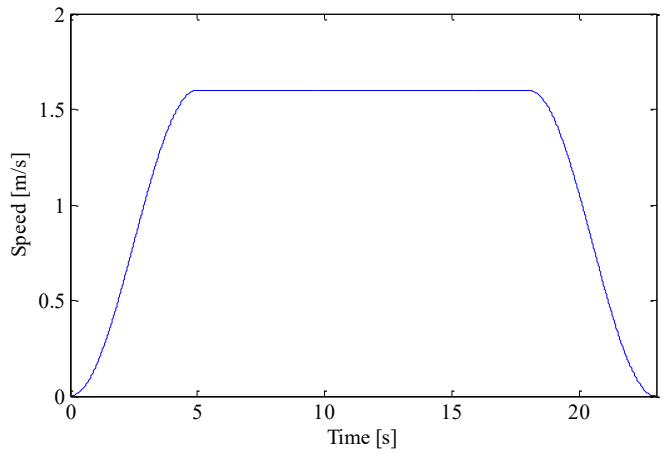


Fig 10 Car speed

The most important vertical excitation, which is also dominant and driving component, is the traction motor torque. The actual motor torque is limited by its power and should be measured by specific torque sensors. Here we use the second law of Newton to represent the motor force as

$$F(t) = \sum m_i \dot{v} \tag{20}$$

where i is the number of mass components in the driving side of traction motor.

7.1.2 Horizontal excitation

The guide rail unevenness excitation is the main factor in horizontal direction that cause vibration. Taking the inevitable manufacturing and assembling error, as well as the impulse result from guide rail joint into consideration, the horizontal deviation can be expressed as

$$\text{dis} = \text{random} + b(t) \tag{21}$$

Since the guide rail joint is distributed evenly in vertical height, it should be transferred into time domain distribution by inverse function solving or if the individual guide rail length is known, the correspondence between time and travel route can be illustrated with a simple diagram as **Figure 11**.

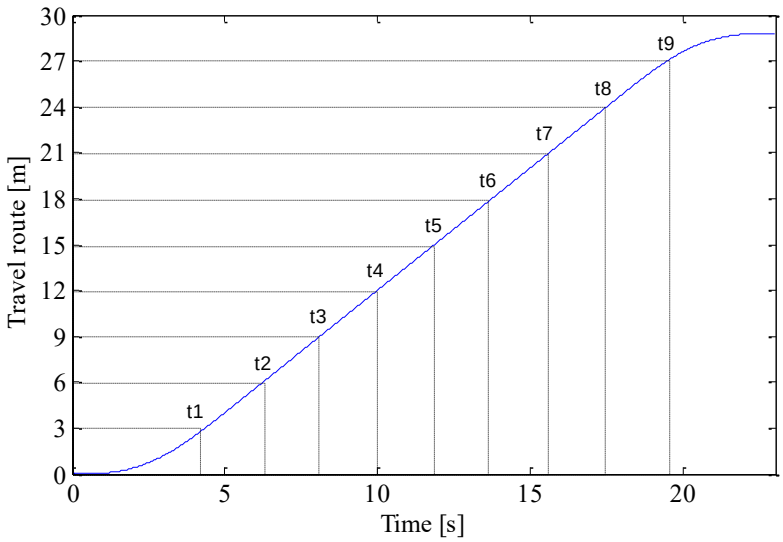


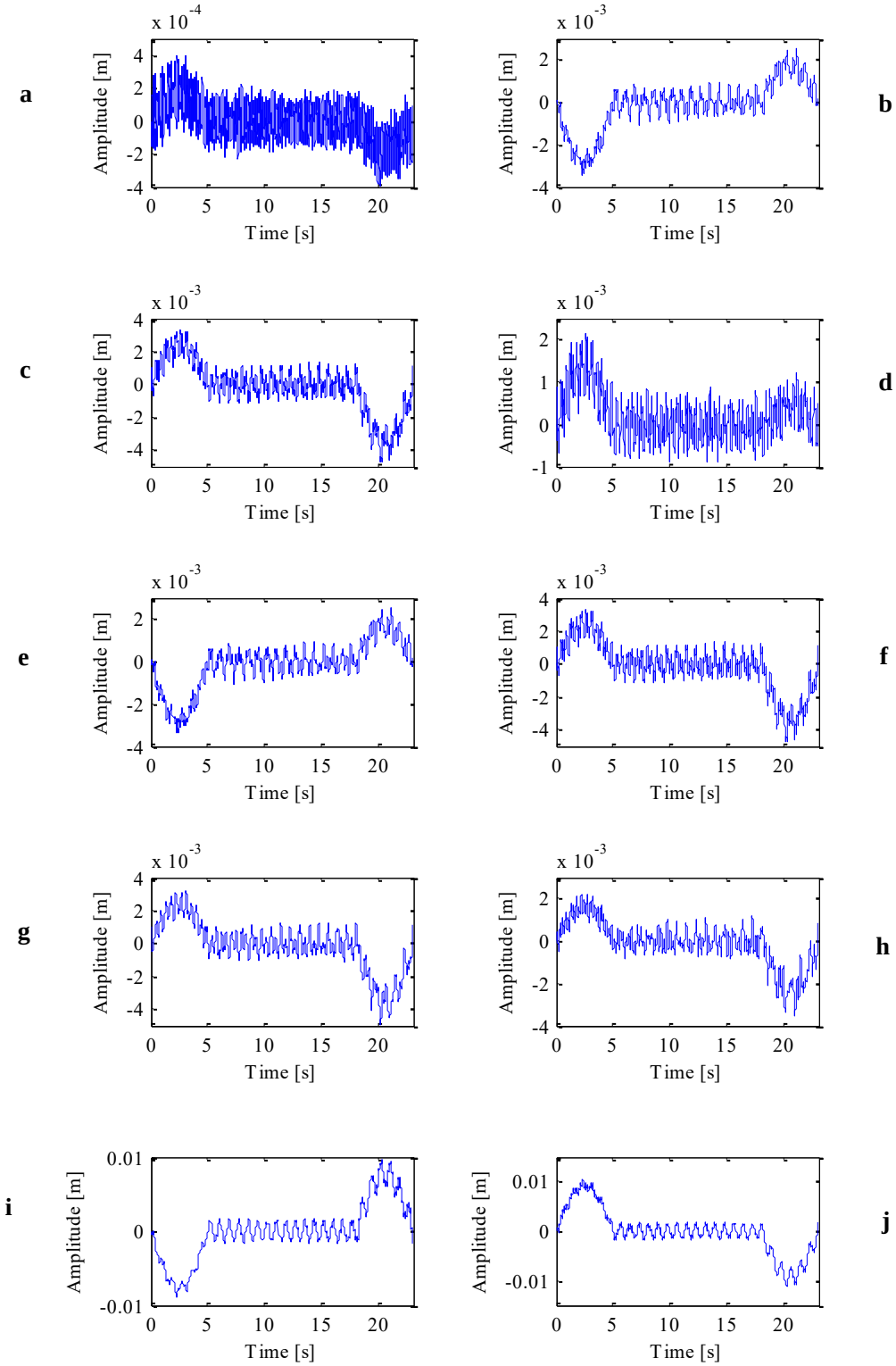
Fig 11 Correspondence of guide rail joint distribution

Thus the guide rail horizontal stimulations can be cast as

$$F_i = -k_z(y_i - \text{dis}) \tag{22}$$

7.2 Vertical vibration analysis

The driving motor force is added in the model to reflect the vibration condition in real world running situation. Given the lift car initial tiny vertical displacement of 0.001m, while other initial values are all set to be 0, the vibration response of each order can be obtained as presented in **Figure 12 (a-q)**.



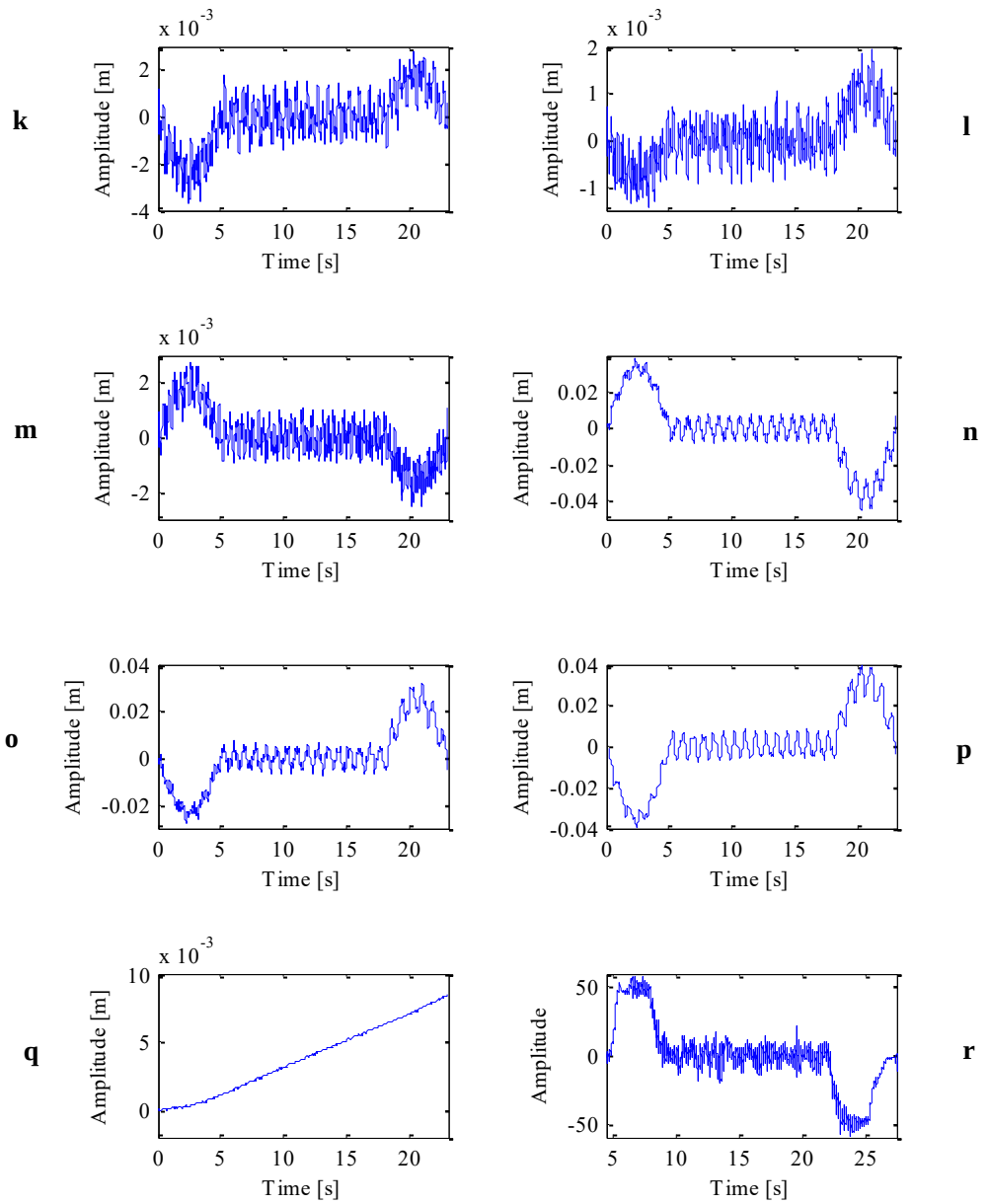
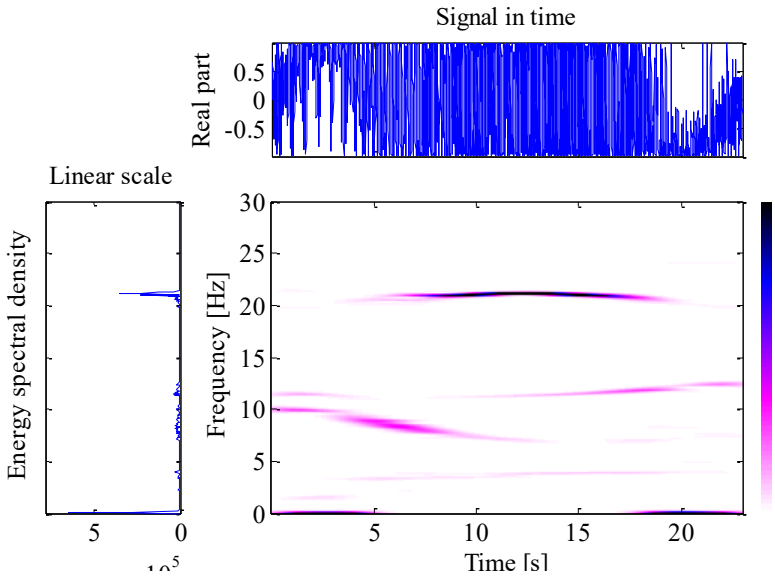
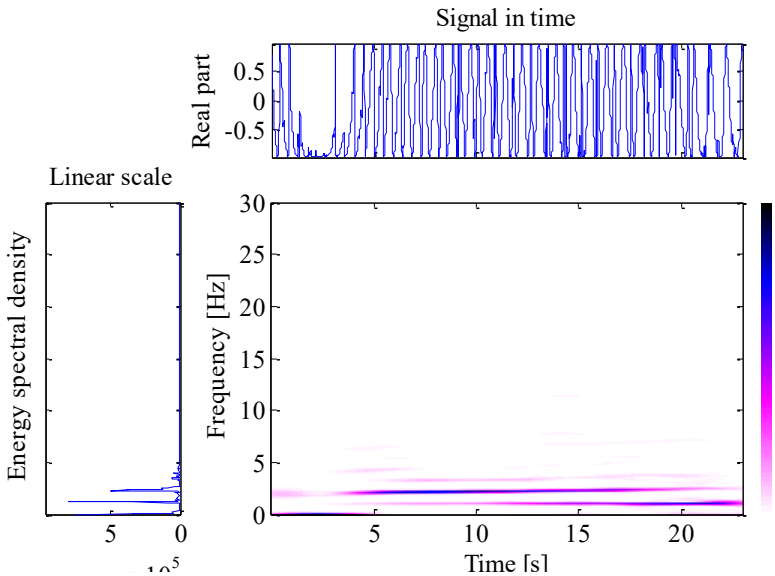


Fig 12 Vertical forced vibration

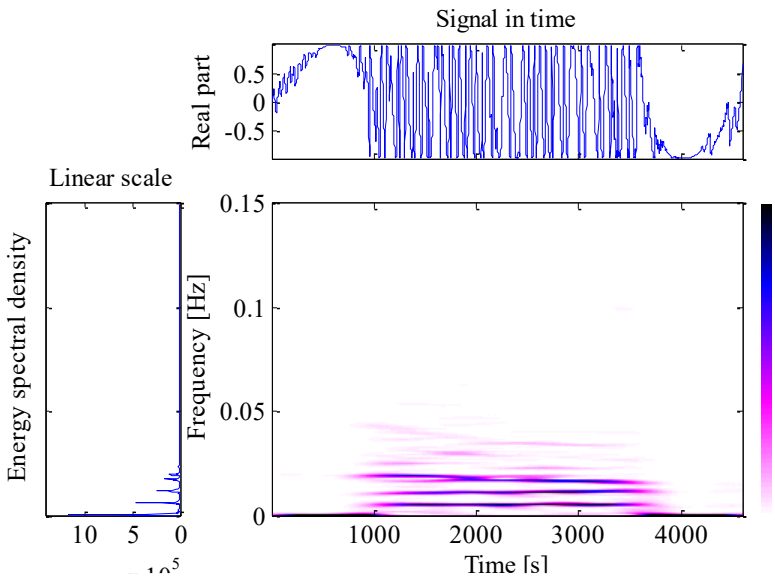
By comparing the simulated forced vibration response waveform with the car vibration waveform perceived by accelerometer in site test (as presented in **Figure 12(r)**), we can easily find the similarity, thus the vertical model is again validated in forced vibration case. Again we analyze the characteristic frequency constitution of main parts forced vibration response, by applying STFT.



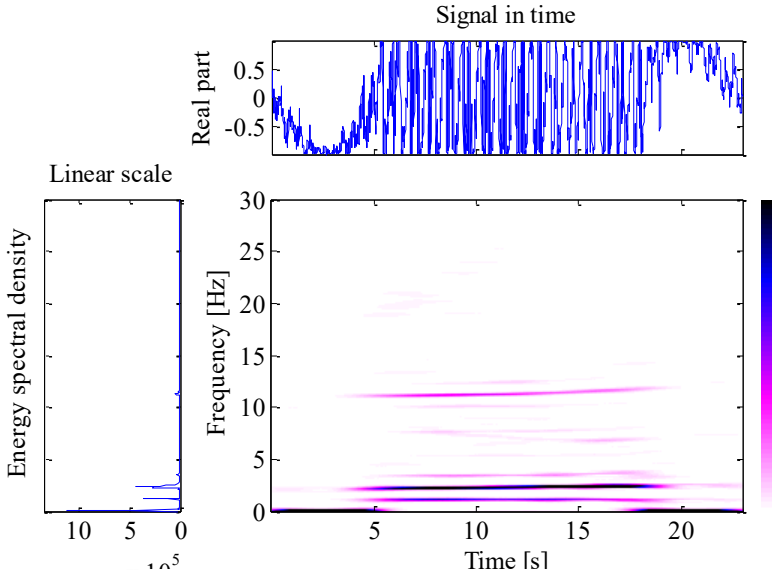
(a) TFR of motor vibration signal (forced vibration)



(b) TFR of car vibration signal (forced vibration)



(c) TFR of rope end (counterweight side) vibration signal (forced vibration)



(d) TFR of rope end (car side) vibration signal (forced vibration)

Fig 13 TFRs of simulated vertical forced vibration

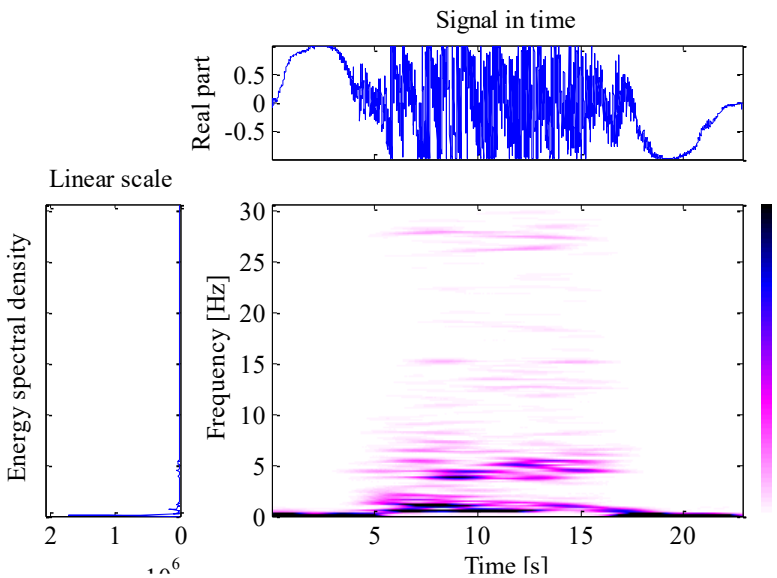


Fig 14 TFR of actual collected car vertical forced vibration

The TFRs of simulated vertical forced response in motor, car and both rope ends are exhibited in Figure 13(a-d), in which the time-varying frequency feature are clear. In comparison with TFRs of free vibration, those high order natural frequencies are missing or with low amplitude, while low order natural frequencies are dominant. It is also consistent with real world case as shown in **Figure 14**. Besides, the motor vibration is of very clear time variant feature which can help identify potential character, while car and rope vibration are of mostly low frequency range component.

7.3 Horizontal vibration analysis

Set the spacing of adjacent guide rail joint equal to floor height 3m, the corresponding moment of impulses can be predicted. The simulated horizontal vibration response are plotted in **Figure 15(a-h)**.

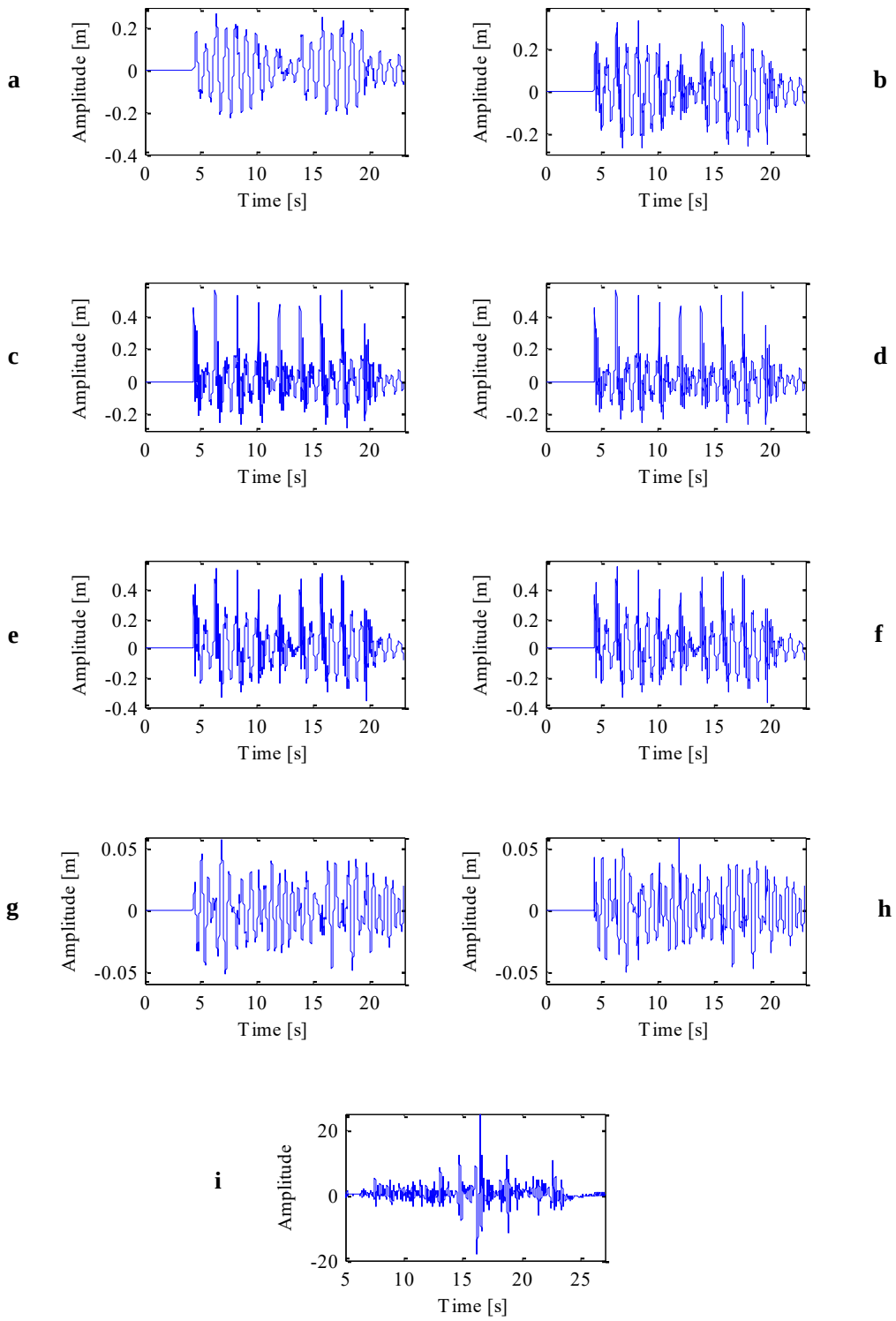


Fig 15 Horizontal forced vibration

For comparison, the in-situ sensor perceived car horizontal vibration is plotted in Figure 15(i). Since the horizontal vibration is weak in amplitude, and are weakened by damping system, the actual sensor perceived horizontal vibration is always featured with impulses result from guide rail joint and assembling error, as obvious in Figure 15(i).

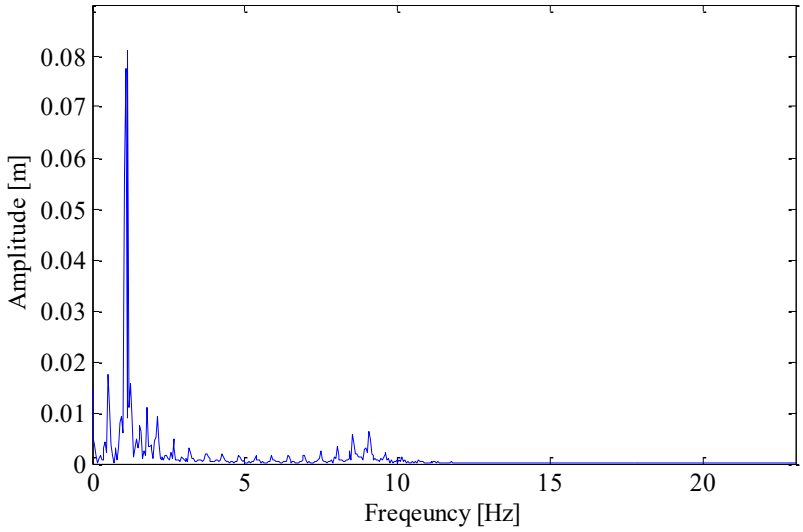


Fig 16 Fourier spectrum of car horizontal forced vibration

Except for the impulses, from the Fourier spectrum in **Figure 16** we can also observe that the frequency constitution is much similar with free vibration case. This can be explained as the unevenness of guide rail is simulated by a gauss wave (when the car speed is higher, the displacement is larger), and the actual value is random. Therefore the vibration form is not changed and the natural frequencies are still visible. In actual use, we recommend the impulses detection in condition evaluation, for the impulses are most likely caused by guide shoe, pulley and/or hoist rope local fault.

7.4 Conclusion

The main external excitation in both vertical and horizontal direction are firstly studied and corresponding external force are added in model to mimic actual forced vibration. In vertical direction, the motor force is undoubtedly the dominant excitation while potential guide rail joint impulses and unevenness are under focus in horizontal direction. Vertical vibration signal waveform clearly reveals the speed up, constant speed and speed down stage, and the TFRs of each order generated by STFT show that the natural frequencies are still dominant in forced vibration case, with extra nonstationarity in amplitude. In horizontal forced vibration signal analysis the unevenness and impulses are added and the corresponding impulses are successfully detected in waveform. The result of above analysis are also validated with actual collected signals, thus the correctness and effect of the proposed model are again demonstrated.

Under the excitation of motor force in vertical direction and guide rail stimulation force in horizontal direction, the elevator is of unique vibration feature. By using the proposed model for excitation simulation and response solving, the waveform and frequency constitution of each component are presented. The above forced vibration analysis in time domain, frequency domain and joint time and frequency domain will also provide reference in actual elevator vibration signal analysis case.

8 CONCLUSION AND FUTURE WORK

By studying the structure and vibration mechanism of the project elevator, both a 17 DOF vertical and 8 DOF horizontal vibration model are constructed based on Lagrange equation and systems of ordinary differential equations. Numerical solving methods including numerical ODE solution on Matlab and Newmark solution are introduced to obtain natural frequencies and vibration responses. The waveform of each order of vibration response are compared with real world collected signal case, and the result validate the correctness of proposed model. Fourier spectrum and short time Fourier transform are used to reveal the frequency constitution and time-variant characteristic frequency structure. The free vibration analysis and forced vibration analysis indicate that low order of natural frequencies are effective in characteristic frequency identification, while impulses and extra frequency components may conceal potential fault feature and deserve much attention in analysis. All these achievements shows that the proposed model is of effectiveness in vibration analysis, fault simulation and characteristic components identification.

To further validate the proposed model and improve the precision, in-situ test are quite necessary. The 2nd site test was conducted on Feb. 6th to overcome the drawbacks of the 1st test, and the corresponding analysis work will be carried out. The 2nd phase of the project mainly focus on measurement, therefore more innovative thinking on collecting useful condition information together with scenarios and tests are expected in the future.

REFERENCE

[1] G. Zhang. Study on elevator dynamic characteristic and monitoring & analysis system development. Master Thesis, Zhejiang University, 2008. (in Chinese)

[2] W. Jin, H. Zhang, Y. Zhen. Modeling and dynamic analysis of elevator. Journal of Machine Design and Research, 1999(3): 53-55. (in Chinese)

[3] Y. Wang. Study on elevator vibration attenuation and noise reduction. Master Thesis, Shanghai Jiao Tong University, 2001. (in Chinese)

[4] D. Yu, J. Yu, B. Chen. KED analysis and dynamic performance optimization of express elevator. Chinese journal of applied mechanics, 1998(2): 41-43. (in Chinese)

[5] P. Lei. Research on elevator traction and vibration. Master Thesis, University of Science and Technology Beijing, 2002. (in Chinese)

[6] X. Gang. Dynamic model and parameter optimization method study for traction elevators. Doctoral Thesis, Zhejiang University, 2005. (in Chinese)

[7] J. Zhang, Q. Yang, Y. Wang. Optimal active control of high speed elevator vibration in time domain. Journal of Zhejiang University of Technology, 2000，28(1): 78-83. (in Chinese)

[8] J. Zhang, Q. Yang. Optimal active control of high speed elevator vibration in frequency domain. Journal of Zhejiang University of Technology, 2000, 28(3): 230-235. (in Chinese)

[9] K. Nai, W Forsythe and R M Goodall. Modelling and simulation of a lift system. IMA International conference on control, Sep 1992.

[10] K Nai, W Forsythe and R M Goodall. Improving ride quality in high-speed elevators. World Elevator, 1997, 45(6), 88-93.

[11] Y. Feng. Active Control of Horizontal Vibrations for High-speed Elevator. Doctoral Thesis, Shanghai Jiao Tong University, 2008. (in Chinese)

[12] W. Fu, C. Zhu. Horizontal vibration modeling and dynamic response analysis of high speed elevator. Journal of Machine Design and Research, 2003, 19(6): 65-67. (in Chinese)

[13] L. Guo, G. Zhang, X. Li. Modeling of elevator cabin-rail coupled dynamic system and its properties. Chinese Journal of Mechanical Engineering, 2007, 43(8): 186-190. (in Chinese)

[14] W. Zhu and G. Xu. Vibration of elevator cable with small bending stiffness. Journal

of Sound and Vibration, 2003, 263(3): 679-699.

[15] G. Forsythe, M. Malcolm, and C. Moler. Computer Methods for Mathematical Computations, Prentice-Hall, New Jersey, 1977.

[16] D. Kahaner, C. Moler, and S. Nash. Numerical Methods and Software, Prentice-Hall, New Jersey, 1989.

[17] J. Dormanz and P. J. Prince. "A family of embedded Runge-Kutta formulae," J. Comp. Appl. Math., Vol. 6, 1980, pp. 19–26.